

Eco-Evolutionary Rescue and Basin Structure in a Roy-Style Predator–Prey Model

Mathbio research notes

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Abstract

We study whether spatial structure can alter indirect evolutionary rescue in a predator–prey model with evolving prey defense. A fixed-defense spatial control did not produce robust predator persistence beyond the well-mixed reference. Introducing prey defense evolution in the well-mixed model increased the predator mortality stress tolerated by the system, consistent with indirect evolutionary rescue through a decline in defense frequency. In the spatial eco-evolutionary model, however, persistence was not organized by a single monotone stress threshold. Multi-horizon, continuation, and basin analyses showed path-dependent dynamics and bistability: at the same mortality stress, predator-persistent and predator-extinct outcomes were both reachable from different initial conditions. A focused basin-boundary scan showed that initial defense frequency and predator abundance influence basin entry. Subsequent ODE–PDE comparison showed that this basin dependence is primarily reaction-dominated: representative PDE solutions remained nearly homogeneous and matched ODE trajectories from the same initial means. A controlled nonlinear trade-off extension recovers the linear compensation branch and supports selected concave, convex, and mixed nonlinear branches, but remains a local structured test rather than a global theorem. These results suggest that spatially extended eco-evolutionary rescue should be analyzed in terms of reachable outcome basins rather than scalar persistence thresholds, while spatial-pattern-mediated rescue is not currently supported in this tested parameterization.

1 Introduction

Evolutionary rescue occurs when adaptive trait or genotype-frequency change allows a population to persist after environmental stress [Gomulkiewicz and Holt, 1995, Bell and Gonzalez, 2009, Bell, 2017]. In the present predator–prey setting, the relevant trait is prey defense. Higher defense reduces predation pressure, but it also carries ecological costs through lower prey growth and lower predator conversion opportunity. Indirect evolutionary rescue of the predator can occur when increased predator mortality lowers predator pressure, selection favors reduced prey defense, and the resulting lower-defense prey state improves predator recovery [Yamamichi and Miner, 2015].

This mechanism cannot be tested in a fixed-defense model. A fixed-defense spatial predator–prey model can ask whether spatial patterning alone creates predator persistence beyond a well-mixed reference, but it cannot test evolutionary rescue because the prey defense state does not change. The eco-evolutionary model therefore introduces a defense frequency q , with $q = 0$ representing fully undefended prey and $q = 1$ representing fully defended prey.

Comparing well-mixed and spatial models is useful because diffusion can alter local predator–prey overlap, transient recovery paths, and the basins of attraction of persistent or extinct states. Predator–prey systems are also a central setting in which rapid evolution and ecological dynamics

can interact on similar timescales [Yoshida et al., 2003, Kasada et al., 2014]. A scalar mortality threshold m_c is a natural first diagnostic: below the threshold the predator persists, while above it the predator fails to persist. This diagnostic is appropriate when the stress response is monotone and when persistence is controlled by one attracting outcome. It can fail in bistable systems, where persistent and extinct states are both reachable at the same stress.

This report distinguishes spatially extended dynamics from genuinely spatial-pattern-mediated dynamics. A PDE can preserve basin-dependent outcomes without those basins being generated by persistent spatial patterning. The main result is therefore two-part: the well-mixed eco-evolutionary model supports indirect evolutionary rescue, and the spatially extended model is not summarized by a single persistence threshold, but the current mechanism diagnosis is reaction-dominated homogeneous multistability rather than spatial-pattern-mediated rescue.

1.1 Relation to previous eco-evolutionary rescue models

This report follows the conceptual structure of indirect evolutionary rescue models in which ecological persistence depends on evolutionary change in another component of the community. Here the evolutionary coordinate is prey defense frequency. Predator mortality stress changes predator pressure; predator pressure changes selection on defense; and the resulting defense state feeds back on prey growth, prey palatability, and predator conversion.

The defense trade-off is intentionally minimal: defended prey have lower growth and lower vulnerability or conversion value than undefended prey. Such defense-growth and defense-palatability trade-offs are enough to create a feedback in which a stressed predator may recover indirectly when prey defense declines [Yamamichi and Miner, 2015]. The form of these trade-offs can qualitatively alter eco-evolutionary predator-prey dynamics [Kasada et al., 2014].

The present model is closest in spirit to indirect evolutionary rescue models in which predator mortality stress changes selection on prey defense. In the well-mixed case, the mechanism matches the qualitative logic proposed by Yamamichi and Miner [2015]: predator decline reduces predation pressure, lower predation pressure favors less-defended prey, and the resulting change in prey state improves predator persistence. The spatial extension studied here asks whether this mechanism remains describable by a mortality threshold once diffusion and spatial initial-condition dependence are introduced.

The contribution of the present report is mathematical and numerical rather than a final biological generalization. It shows that adding spatial dynamics changes the rescue question from a scalar threshold comparison to a basin-structured dynamical problem in the tested parameterization.

2 Model formulation

The eco-evolutionary variables are total prey density n , predator density w , prey defense frequency q , free space z , and predator mortality stress s . Free space is

$$z = \kappa^{-1} - n - w.$$

The defense trade-off functions are

$$r(q) = r_u(1 - q) + r_v q,$$

$$a(q) = a_u(1 - q) + a_v q,$$

$$b(q) = b_u(1 - q) + b_v q.$$

The well-mixed eco-evolutionary ODE is

$$\frac{dn}{dt} = n[r(q)z - \xi - a(q)w], \quad (1)$$

$$\frac{dw}{dt} = w[b(q)nz - (m + s) - \mu w], \quad (2)$$

$$\frac{dq}{dt} = \nu q(1 - q)[(r_v - r_u)z - (a_v - a_u)w]. \quad (3)$$

The spatial extension uses the same local reaction terms and adds diffusion:

$$\partial_t n = D_n \nabla^2 n + n[r(q)z - \xi - a(q)w], \quad (4)$$

$$\partial_t w = D_w \nabla^2 w + w[b(q)nz - (m + s) - \mu w], \quad (5)$$

$$\partial_t q = D_q \nabla^2 q + \nu q(1 - q)[(r_v - r_u)z - (a_v - a_u)w]. \quad (6)$$

The spatial simulations use zero-flux Neumann boundary conditions.

Roy eco-evolutionary rescue model variables

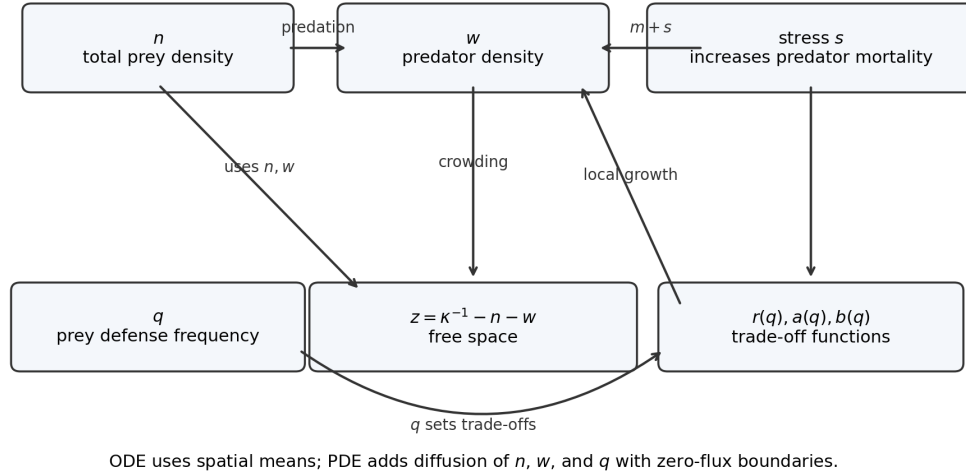


Figure 1: Eco-evolutionary model variables and trade-off links. Arrows indicate that defense frequency q sets the growth, attack, and conversion functions $r(q)$, $a(q)$, and $b(q)$; prey and predator densities determine free space $z = \kappa^{-1} - n - w$; and mortality stress s increases predator mortality.

3 Numerical questions and diagnostic criteria

The first numerical question was whether prey defense evolution increases the predator mortality stress tolerated by the well-mixed system. This was measured by comparing the no-evolution and evolution thresholds

$$m_c^{ODE, \text{no evo}}, \quad m_c^{ODE, \text{evo}},$$

Table 1: Parameter values used in the eco-evolutionary tests. The conversion values $b_u = 0.08$ and $b_v = 0.02$ were used to obtain an interior defense-frequency baseline.

Symbol	Value	Description
κ	0.15	free-space scale
ξ	0.55	prey loss term
r_u	1	undefended prey growth
r_v	0.65	defended prey growth
a_u	1	undefended predation coefficient
a_v	0.35	defended predation coefficient
b_u	0.08	undefended conversion coefficient
b_v	0.02	defended conversion coefficient
m	0.1	baseline predator mortality
μ	0.2	predator density dependence
ν	0.05	evolutionary rate
D_n	0.01	prey diffusion
D_w	0.01	predator diffusion
D_q	0.005	defense-frequency diffusion

and their difference

$$\Delta_{evo}^{ODE} = m_c^{ODE, \text{evo}} - m_c^{ODE, \text{no evo}}.$$

A positive value indicates that prey evolution expands the predator persistence range in the well-mixed model.

The second numerical question was whether adding spatial diffusion changes the same rescue effect. A scalar PDE threshold was initially tested because it gives a direct comparison with the ODE threshold when persistence changes monotonically with stress. The later spatial diagnostics were needed because the apparent PDE boundary was sensitive to the tail window, integration horizon, and tail-slope rule. Multi-horizon analysis showed long transients near the apparent boundary, continuation experiments showed direction-dependent outcomes, and basin scans showed that different initial conditions can reach different outcomes at the same stress.

The outcome classes used in the spatial interpretation are scientific rather than procedural. A persistent outcome has a predator density tail that remains positive and steady enough to support persistence. An extinct outcome has predator density near zero. A transient outcome has not settled clearly into either class at the tested horizon. A bistable stress is one where both persistent and extinct outcomes are reachable from different initial states or continuation paths.

3.1 Numerical protocol

Well-mixed ODE thresholds were estimated by stress continuation and bisection, with predator persistence assessed from tail behavior. PDE scalar thresholds were used only as preliminary screening quantities because a scalar threshold is meaningful only if stress response is effectively monotone.

Spatial simulations used zero-flux boundaries and diffusion in n , w , and q . Initial PDE persistence calls were based on final-time and tail-window behavior, but sensitivity to tail window, integration horizon, and tail-slope rule motivated multi-horizon diagnostics. Continuation tests then used final states from neighboring stress values as initial states to test path dependence. Basin scans first varied broad initial-condition families and later varied q_0 and w_0/w_{baseline} directly.

PDE outcomes were interpreted as persistent, extinct, transient, or bistable using tail behavior, steady-state residual checks, and whether multiple outcomes were reachable at the same stress.

4 Fixed-defense spatial model as a control

Before introducing prey evolution, the fixed-defense spatial model was used as a control for spatial patterning alone. The question was whether spatial heterogeneity could create predator persistence beyond the well-mixed reference when prey defense was held fixed.

The fixed-defense spatial model did not produce robust ODE-extinct/PDE-persistent predator rescue. The strongest weak candidate occurred at $D_w/D_u = 150$, but further grid and mechanism checks classified it as a transient or numerical candidate rather than a robust spatial rescue mechanism. This control establishes that fixed spatial patterning alone was insufficient for predator rescue in the tested model.

5 Indirect evolutionary rescue in the well-mixed model

The well-mixed eco-evolutionary model gave the clean positive result. The no-evolution predator mortality stress threshold was

$$m_c^{ODE, \text{no evo}} = 0.069448242,$$

whereas the evolving-defense threshold was

$$m_c^{ODE, \text{evo}} = 0.16486816.$$

Thus

$$\Delta_{evo}^{ODE} = 0.095419922.$$

In the rescue window, defense frequency declined from approximately 0.6726 to approximately 0.3336.

Mechanistically, increased predator mortality reduces predator pressure. Lower predator pressure makes defense less favorable because the growth cost of defense is no longer offset by strong predation avoidance. As q declines, prey growth and predator conversion opportunity increase, allowing predator persistence over a larger stress interval. This is the cleanest evidence for indirect evolutionary rescue in the model sequence and follows the same qualitative feedback identified by [Yamamichi and Miner \[2015\]](#).

Table 2: Threshold quantities from the ODE rescue analysis and preliminary PDE screening. The PDE entries are early screening quantities; later diagnostics show that they should not be treated as final scalar thresholds for the spatial system.

Quantity	Value	Interpretation
$m_c^{ODE, \text{no evo}}$	0.0694482422	ODE reference
$m_c^{ODE, \text{evo}}$	0.164868164	ODE reference
Δ_{evo}^{ODE}	0.0954199219	ODE evolutionary rescue effect
representative $m_c^{PDE, \text{no evo}}$	0.06921875	historical screening quantity
representative $m_c^{PDE, \text{evo}}$	0.11765625	historical screening quantity

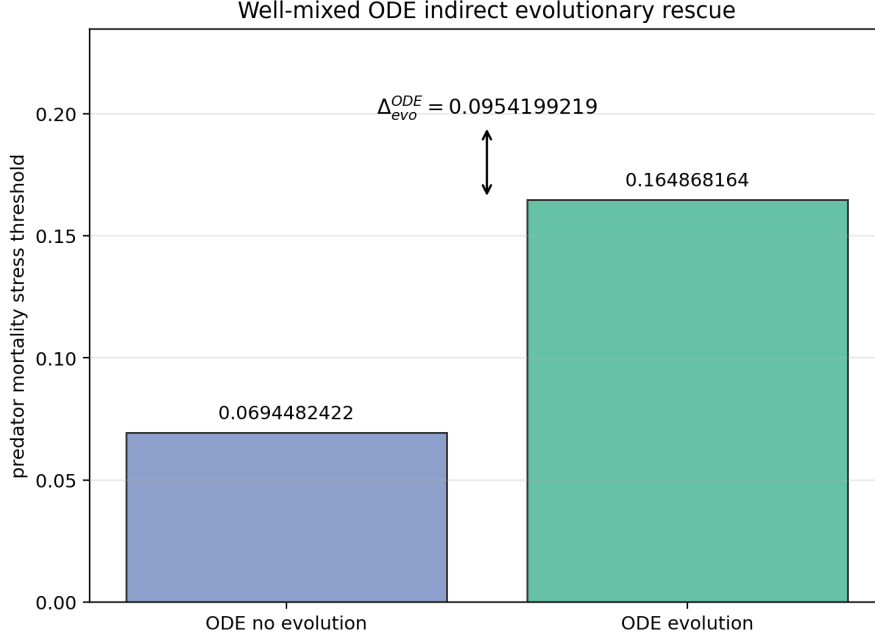


Figure 2: Well-mixed ODE threshold comparison. The no-evolution threshold is $m_c^{ODE, \text{no evo}} = 0.069448242$, the evolving-defense threshold is $m_c^{ODE, \text{evo}} = 0.16486816$, and the evolutionary rescue effect is $\Delta_{evo}^{ODE} = 0.095419922$.

6 Spatial eco-evolutionary model

The spatial eco-evolutionary model adds diffusion of n , w , and q to the same local eco-evolutionary reactions. The first scalar-threshold screen suggested that the PDE tolerated less mortality stress than the well-mixed evolving system. The representative screening values were $m_c^{PDE, \text{no evo}} = 0.06921875$ and $m_c^{PDE, \text{evo}} = 0.11765625$. Because $0.11765625 < 0.16486816$, this preliminary comparison suggested a lower PDE-evolution persistence boundary than the ODE-evolution threshold.

That comparison was useful as a warning signal, but it was not a stable final summary of the spatial system. The spatial model showed sensitivity to persistence criteria, long transients, and path-dependent outcomes. These features change the scientific question from “What is the PDE threshold?” to “Which persistent, extinct, or transient outcomes are reachable at a given stress?”

7 Evidence for path dependence and bistability

The spatial diagnostics proceeded through four conceptual phases. First, persistence classification was found to depend on the tail window and integration horizon. Second, longer multi-horizon analysis showed that several stresses near the apparent boundary remained transient rather than cleanly persistent or extinct. Third, continuation experiments showed direction-dependent outcomes: upward continuation could maintain a predator-persistent branch at stresses where downward continuation from high stress could remain extinct or transient. Finally, initial-condition basin scans confirmed that predator-persistent and predator-extinct outcomes can both be reached at the same stress.

Continuation data illustrate the path dependence. At stress 0.16486816, upward continuation produced a persistent steady outcome, whereas downward continuation produced an extinct steady

Diagnostics showing why a scalar PDE threshold is insufficient



The spatial PDE response is path-dependent, so scalar threshold language is insufficient.

Figure 3: Diagnostic sequence showing why scalar PDE threshold language failed. Tail-window sensitivity, long transients, continuation dependence, and basin scans together showed that the spatial system is organized by path-dependent outcomes rather than one monotone persistence boundary.

outcome. At stress 0.175, the same contrast persisted. These paired outcomes show that the spatial model cannot be represented by one monotone persistence curve.

The broader basin-regime map identifies two types of stress regimes:

- mixed persistent/transient regimes at 0.141262205 and 0.15;
- bistable persistent/extinct regimes at 0.1584375, 0.16486816, and 0.175.

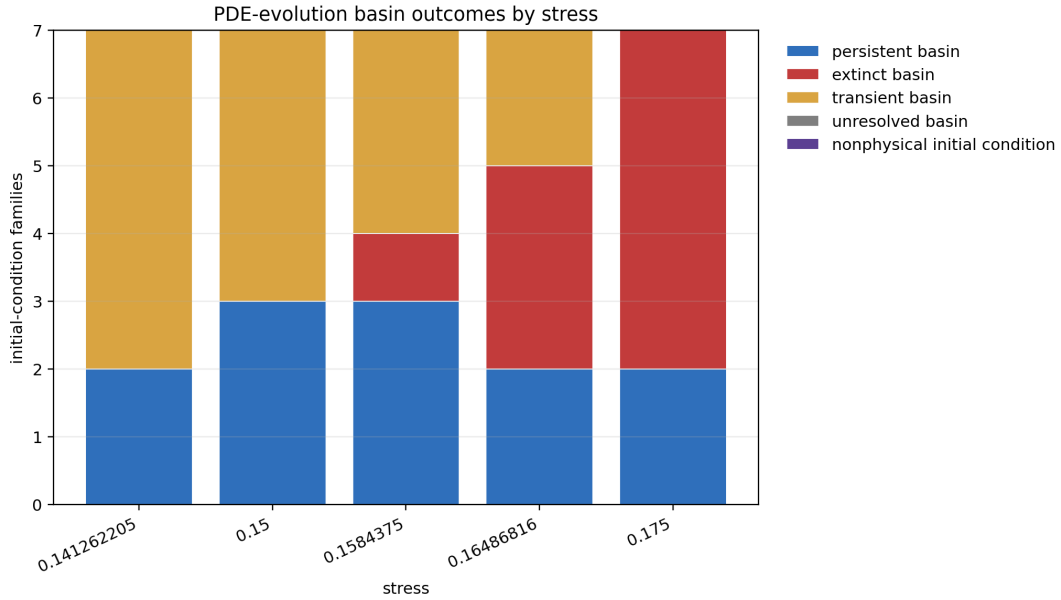


Figure 4: Basin outcomes across initial-condition families. Each stacked bar counts initial-condition families by basin label at a fixed stress; coexistence of persistent and extinct outcomes in the same bar indicates bistability.

Table 3: Stress-regime map from the bistability analysis.

Stress	Regime
0.141262205	persistent_transient_mixed
0.15	persistent_transient_mixed
0.1584375	bistable_persistent_extinct
0.16486816	bistable_persistent_extinct
0.175	bistable_persistent_extinct

8 Basin-boundary dependence on q_0 and w_0

The focused basin-boundary scan varied initial defense frequency q_0 and initial predator abundance scale w_0/w_{baseline} at two stresses inside the bistable interval. At stress 0.1584375, the q_0 - w_0 grid gave 20 persistent, 7 extinct, and 43 transient outcomes. At stress 0.16486816, it gave 14 persistent, 21 extinct, and 35 transient outcomes. No unresolved or nonphysical initial-condition outcomes were recorded in these two focused grids.

The scan shows that basin entry is two-dimensional. Low-to-intermediate q_0 values tend to favor predator-persistent outcomes, so the initial defense state is a major basin-entry coordinate. Predator initial abundance modulates the outcome within that q_0 range. Transient outcomes remain common, which means the separatrix is not yet fully resolved at the tested horizon.

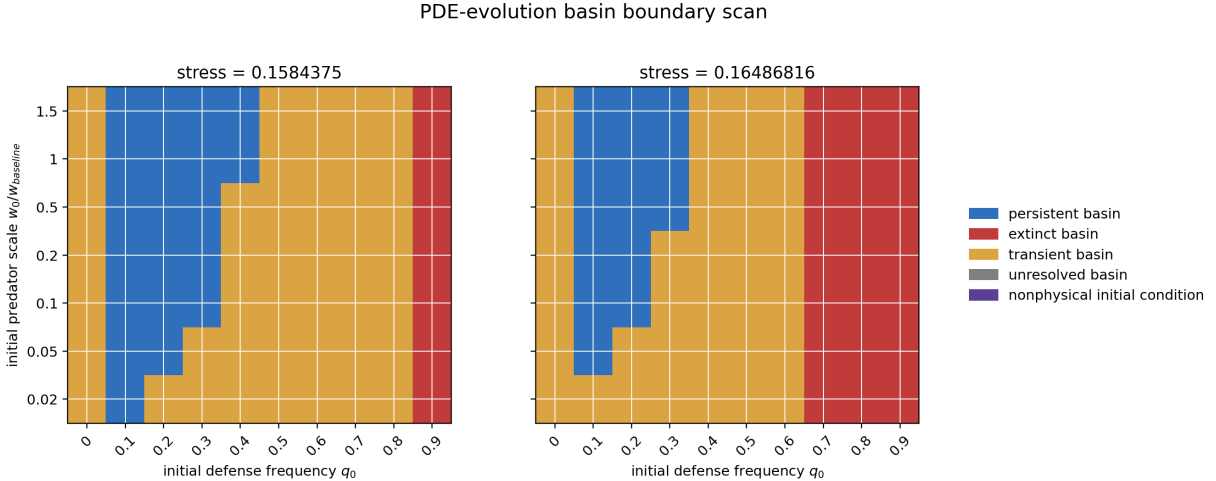


Figure 5: Basin-boundary scan over initial defense frequency q_0 and predator abundance scale w_0/w_{baseline} at stresses $s = 0.1584375$ and $s = 0.16486816$. Colors encode basin labels for persistent, extinct, transient, unresolved, or nonphysical outcomes.

Table 4: Basin-boundary counts from the q_0 - w_0 scan.

Stress	Persistent	Extinct	Transient	Unresolved	Nonphysical	Regime
0.1584375	20	7	43	0	0	bistable_persistent_extinct
0.16486816	14	21	35	0	0	bistable_persistent_extinct

Table 4 highlights the stress-dependent basin balance. At $s = 0.1584375$, persistent and extinct

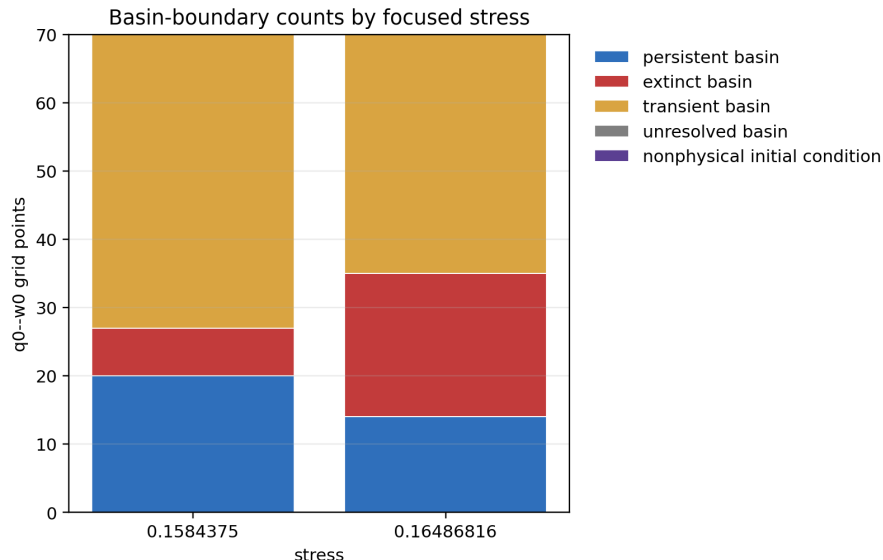


Figure 6: Outcome counts in the focused q_0 – w_0 basin-boundary scan. Transient outcomes remain common at both stresses, so the basin boundary is interpretable but the separatrix is not fully resolved at the tested horizon.

basins both occur, but transient outcomes dominate the grid. At $s = 0.16486816$, extinct outcomes become more common, yet persistent outcomes remain reachable. Therefore the basin boundary is interpretable but not sharply resolved at the current horizon.

9 Mechanism diagnosis: homogeneous reaction dynamics versus spatial patterning

The basin-boundary scan established that the spatially extended PDE preserves multiple reachable outcomes, but it did not by itself identify whether spatial patterning generates those outcomes. A subsequent matched ODE-PDE mechanism test compared representative PDE spatial means with ODE trajectories from the same initial means, quantified representative spatial variance, compared ODE and PDE basin labels on the same q_0 – w_0 grid, and ran targeted perturbation-sensitivity tests.

The mechanism diagnosis was **reaction_dominated_homogeneous_multistability**. Representative ODE-PDE classifications agreed in 3 of 3 cases, basin-grid labels agreed for 90% of 140 q_0 – w_0 points, all 14 disagreements involved transient labels, direct persistent/extinct disagreements were zero, final spatial CV values were very small, and perturbation tests produced no outcome changes. Thus the current evidence indicates that the observed PDE basin structure is mostly inherited from the homogeneous eco-evolutionary reaction dynamics rather than generated by persistent spatial patterning.

The follow-up ODE analysis therefore treats the homogeneous reaction system as the next object of study. The representative ODE trajectories track predator density, defense frequency, the selection gradient

$$(r_v - r_u)z - (a_v - a_u)w,$$

and the predator growth factor

$$b(q)nz - (m + s) - \mu w.$$

The ODE q_0 - w_0 basin map reproduces persistent, extinct, and transient regions on the same focused grid, and the numerical equilibrium scan gives parameter-specific evidence for multiple homogeneous attractors. These diagnostics support the conditional interpretation that basin-dependent rescue is controlled by homogeneous eco-evolutionary feedbacks under the tested trade-off values.

The equilibrium scan suggests that the persistent homogeneous branch compensates increased predator mortality by lowering the equilibrium defense frequency q^* , while n^* and w^* remain close to the same positive values over the tested stress range. This provides a reaction-level explanation for the rescue mechanism and for why the spatial PDE can inherit basin-dependent outcomes without persistent spatial patterning.

9.1 Homogeneous compensation branch

The homogeneous compensation branch can be derived directly from the interior ODE equilibrium equations. Let $c = (r_v - r_u)/(a_v - a_u)$. The selection-gradient condition gives $w = cz$. With the linear trade-offs, $r(q) - ca(q) = r_u - ca_u$, so

$$z^* = \frac{\xi}{r_u - ca_u}, \quad w^* = cz^*, \quad n^* = \kappa^{-1} - z^* - w^*.$$

The predator balance then determines the required conversion value

$$b_{\text{req}}(s) = \frac{m + s + \mu w^*}{n^* z^*},$$

and the conversion trade-off gives

$$q^*(s) = \frac{b_{\text{req}}(s) - b_u}{b_v - b_u}.$$

For the current parameterization, this analytic branch matches the numerical stable persistent equilibria to roundoff precision, with maximum $|\Delta q^*| < 10^{-14}$. Along the branch, q^* decreases as stress increases while n^* and w^* remain approximately fixed. A local structured trade-off check found the compensation branch in 72 of 81 nearby parameter sets, so the mechanism is supported locally but should not be treated as globally general.

Interior compensation branch. For the linear trade-off ODE, an interior persistent equilibrium with $n^* > 0$, $w^* > 0$, and $0 < q^* < 1$ must satisfy

$$w^* = cz^*, \quad c = \frac{r_v - r_u}{a_v - a_u}.$$

When $c > 0$, $r_u - ca_u > 0$, and the resulting

$$z^* = \frac{\xi}{r_u - ca_u}, \quad w^* = cz^*, \quad n^* = \kappa^{-1} - z^* - w^*$$

are positive, the predator equation determines

$$q^*(s) = \frac{(m + s + \mu w^*)/(n^* z^*) - b_u}{b_v - b_u}.$$

The compensation branch is interior for stresses satisfying $0 < q^*(s) < 1$. For the current parameterization, this analytic branch matches the numerical stable persistent equilibria up to machine

precision. The analytic Jacobian is locally stable at the tested target stresses, and the local condition grid identifies a nontrivial stable region. These are conditional results for the tested linear trade-offs, not a global theorem for all defense trade-off forms.

The local stability of the interior compensation branch was also evaluated using the Routh–Hurwitz conditions for the cubic characteristic polynomial of the analytic Jacobian. For a polynomial $\lambda^3 + A_1\lambda^2 + A_2\lambda + A_3$, local stability requires $A_1 > 0$, $A_2 > 0$, $A_3 > 0$, and $A_1A_2 > A_3$. In the current parameterization, these inequalities hold at the target stresses along the compensation branch and agree with the analytic eigenvalue calculation (Figures 13 and 14).

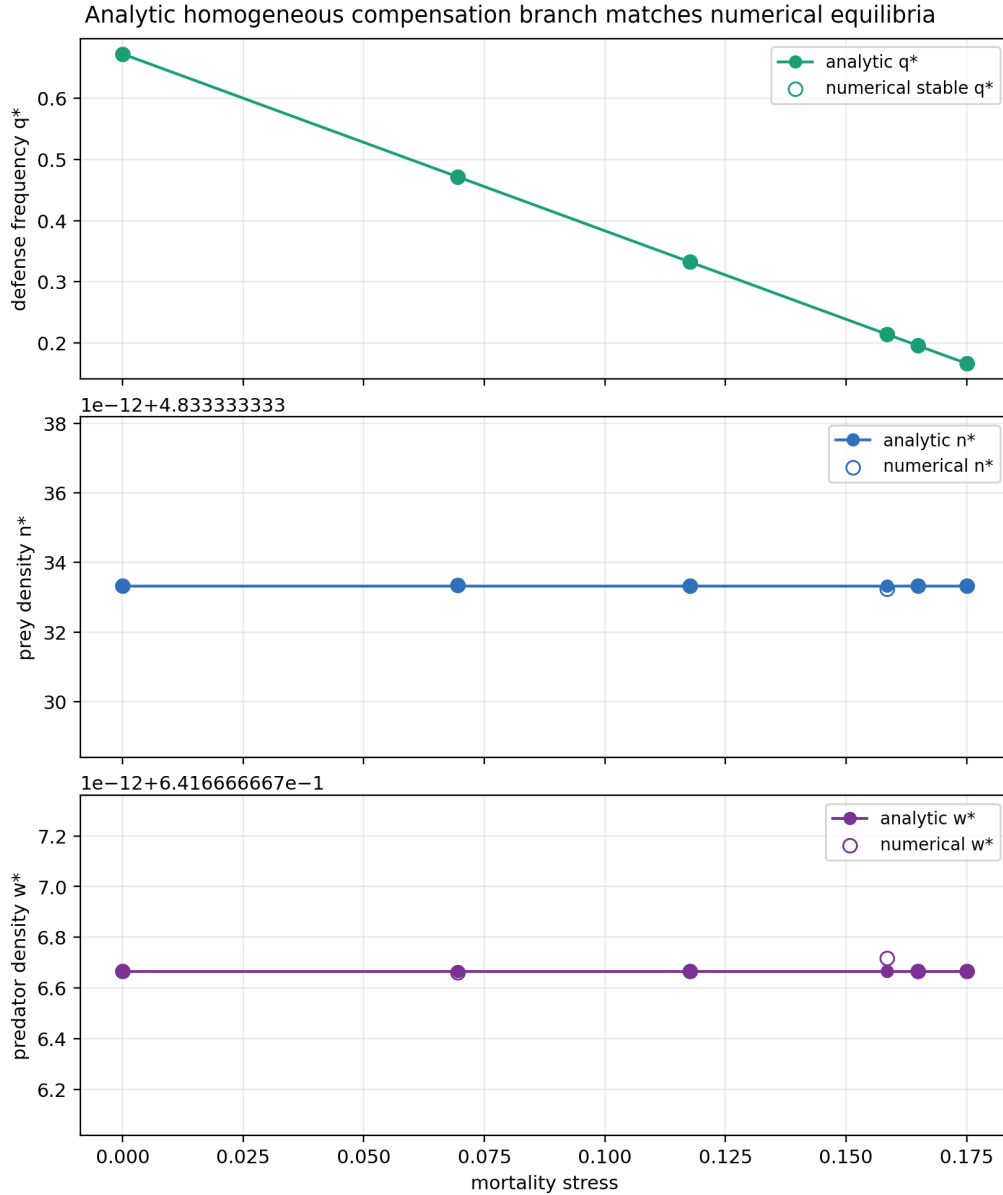


Figure 7: Analytic homogeneous compensation branch compared with the existing numerical stable persistent equilibria. The branch keeps n^* and w^* nearly fixed while $q^*(s)$ declines with stress.

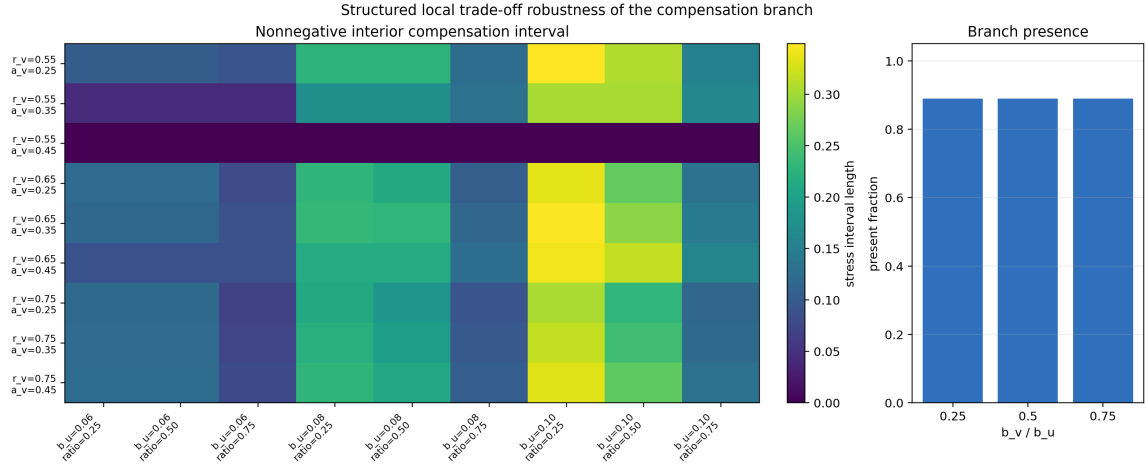
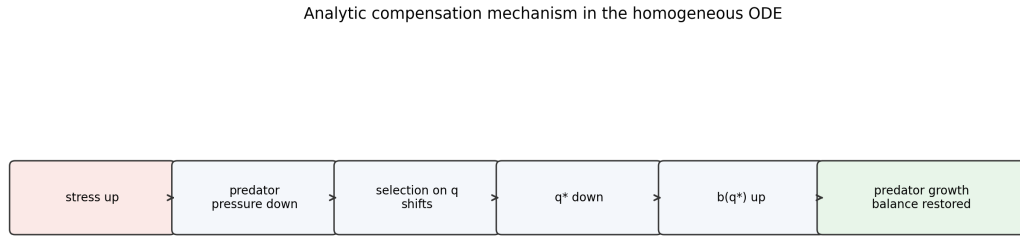


Figure 8: Local structured trade-off robustness check for the analytic compensation branch. Interval lengths are local diagnostics, not a claim of global robustness.



Homogeneous reaction-level compensation: no persistent spatial patterning is required for this branch.

Figure 9: Conceptual homogeneous compensation mechanism: increasing stress lowers predator pressure, shifts selection on q , lowers q^* , increases $b(q^*)$, and restores predator growth balance along the persistent branch.

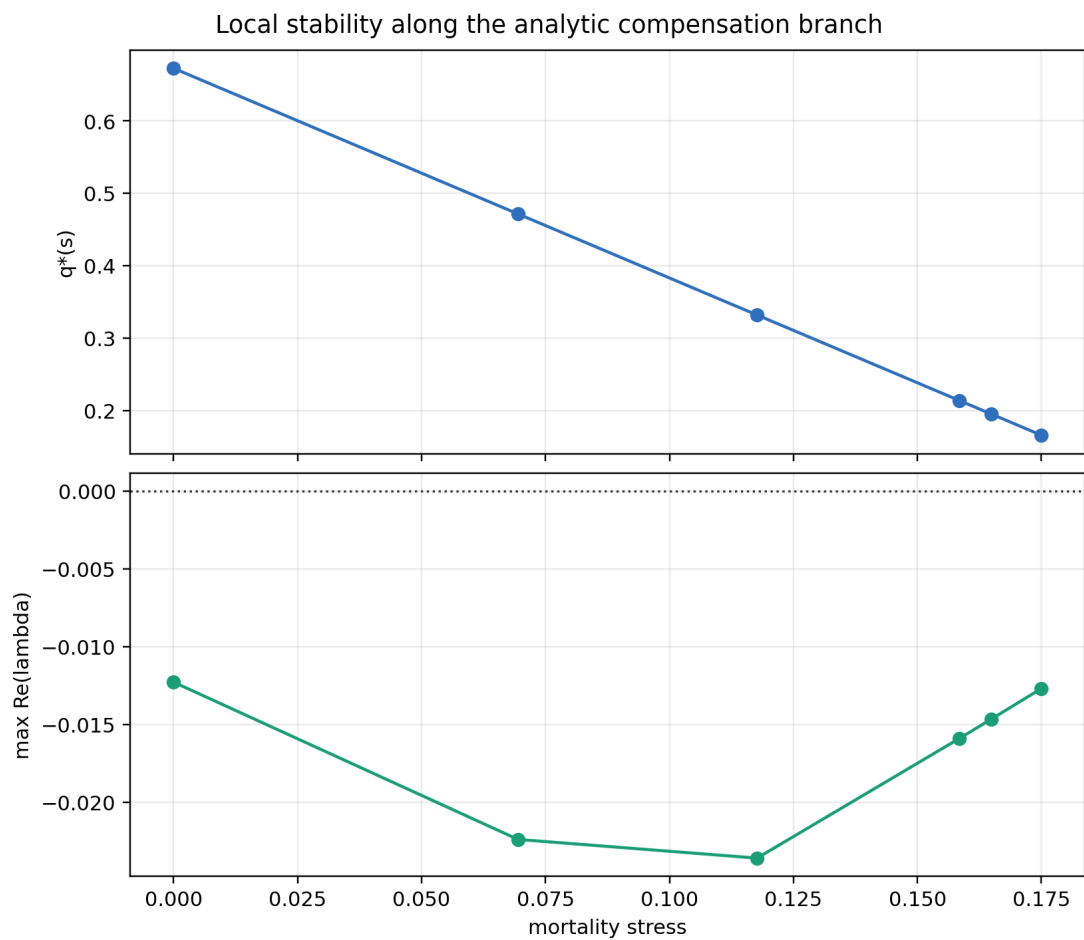


Figure 10: Local stability diagnostics along the current analytic compensation branch. The maximum real part of the analytic Jacobian eigenvalues remains negative at the tested stresses.

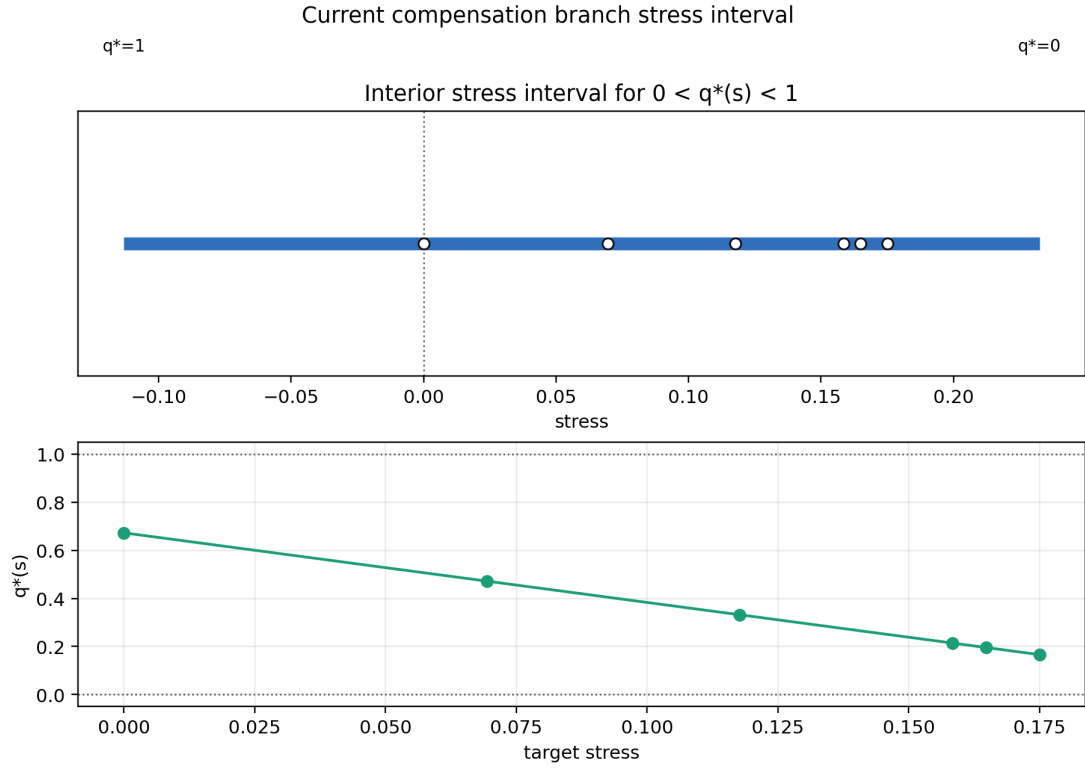


Figure 11: Interior stress interval for the current compensation branch. The branch remains feasible while $0 < q^*(s) < 1$; the target stresses used here lie inside this interval.

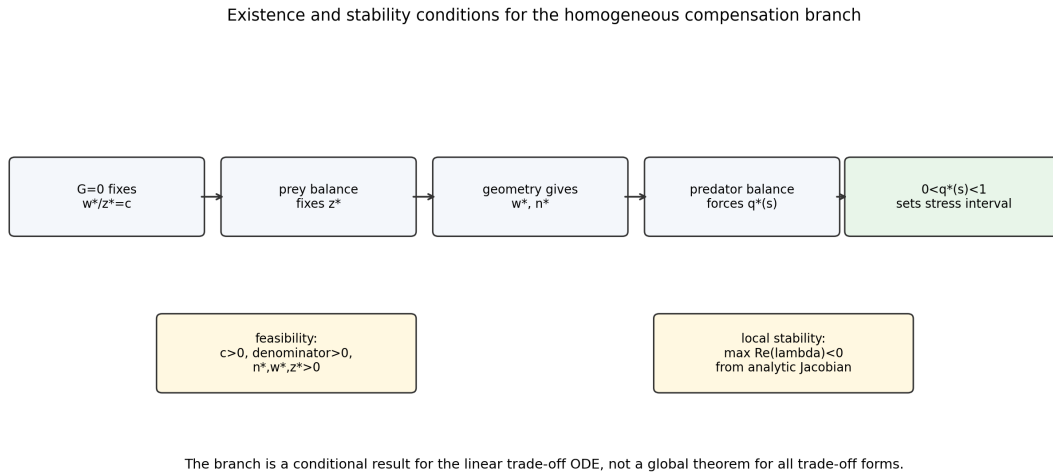


Figure 12: Schematic of the compensation-branch conditions. The selection-gradient condition fixes w^*/z^* , the prey equation fixes z^*, w^*, n^* , and the predator equation determines the stress-dependent $q^*(s)$.

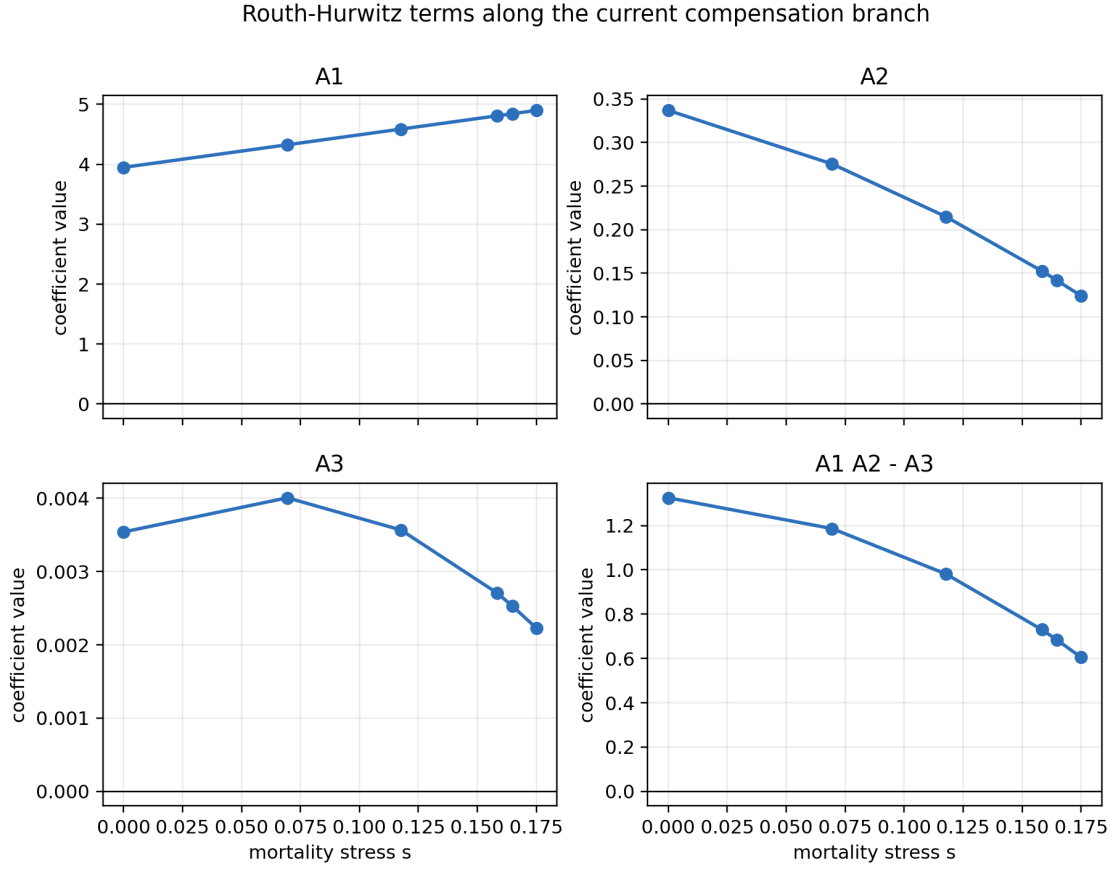


Figure 13: Routh-Hurwitz coefficient terms along the current compensation branch. All terms remain positive at the target stresses.

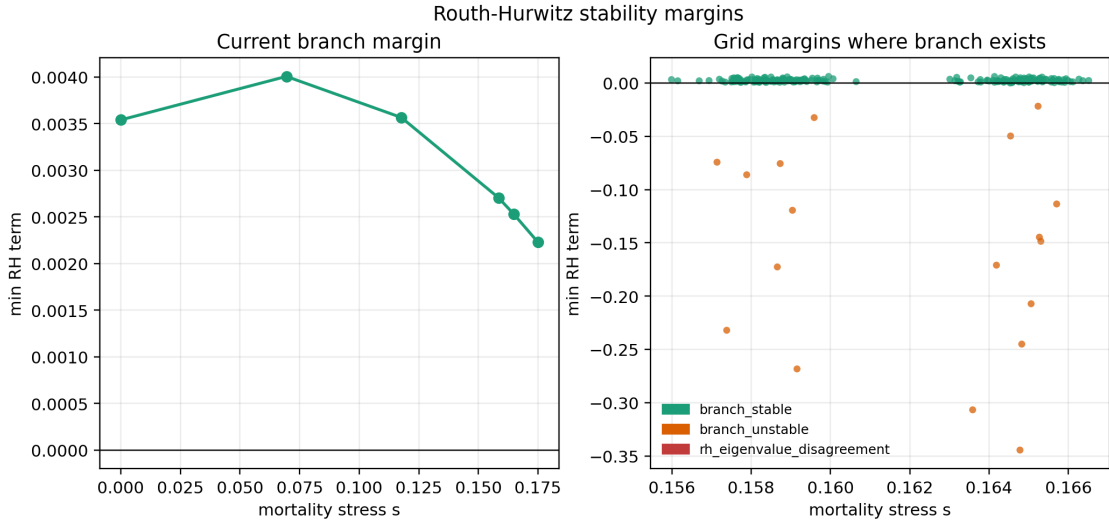


Figure 14: Routh-Hurwitz stability margins for the current branch and the structured local condition grid. Positive margins indicate local stability under the cubic Routh-Hurwitz inequalities.

9.2 Spatial stability and finite-amplitude heterogeneous perturbations

The homogeneous ODE compensation branch is also a homogeneous steady-state branch of the spatial PDE. Linear perturbations around this branch decompose into Neumann spatial modes with matrices

$$J_F(U^*) - \lambda_{mn}D,$$

where $J_F(U^*)$ is the reaction Jacobian and

$$D = \text{diag}(D_n, D_w, D_q).$$

For the current diffusion coefficients, all tested nonzero Neumann modes were stable across the target stresses, and the controlled diffusion-ratio grid showed no linear spatial instability in the tested range (Figures 21–24).

Linear stability does not rule out finite-amplitude basin effects, so targeted nonlinear PDE tests used localized predator patches, localized defense patches, sinusoidal modes, random heterogeneity, and basin-boundary heterogeneity. These tests were compared with matched homogeneous controls. The final decision was spatial stability of the homogeneous branch with finite-amplitude basin effects: two local-defense perturbations from basin-boundary mean states changed the finite-horizon basin label from persistent to transient, while final spatial CV values decayed below threshold and no sustained spatial pattern persisted. Thus the finite-amplitude result qualifies the reaction-dominated interpretation near basin boundaries, but it does not support a persistent spatial-pattern-mediated rescue mechanism in the tested setup (Figures 25–29).

Long-horizon follow-up showed that the finite-amplitude heterogeneous basin changes resolve to the homogeneous-control outcome. The two basin-changing local-defense perturbations reproduced the transient label at $T = 1600$, but both matched their homogeneous-control persistent basin by $T = 3200$ and $T = 6400$, with final spatial CV values below threshold. This clarifies that the PR #23 basin-changing local-defense perturbations were finite-horizon transients rather than persistent basin changes or spatial-pattern-mediated outcomes (Figures 30–33).

9.3 Extension to nonlinear trade-off forms

The compensation-branch calculation also extends to endpoint-preserving nonlinear trade-offs of the form $r(q) = r_u + (r_v - r_u)q^{\gamma_r}$, $a(q) = a_u + (a_v - a_u)q^{\gamma_a}$, and $b(q) = b_u + (b_v - b_u)q^{\gamma_b}$. In this case the branch is parameterized by q , with

$$c(q) = \frac{r'(q)}{a'(q)}, \quad z(q) = \frac{\xi}{r(q) - a(q)c(q)}, \quad w(q) = c(q)z(q), \quad n(q) = \kappa^{-1} - z(q) - w(q),$$

and

$$s(q) = b(q)n(q)z(q) - m - \mu w(q).$$

The linear case is recovered when all exponents are one. In the controlled shape grid with $\gamma \in \{0.5, 1, 2\}$, 11 of 27 shape combinations were stable at all target stresses, 11 were stable at some target stresses, 4 had no feasible target branch, and 1 was unresolved. Selected PDE spatial-mode checks found no nonzero-mode instability, and targeted nonlinear non-homogeneous PDE tests produced one finite-horizon all-convex basin-boundary transient that resolved at the longer horizon. The final label for this controlled extension is `nonlinear_tradeoff_compensation_supported`, with the caveat that nonlinear shape effects remain local and parameter-dependent rather than globally established (Figures 15–20).

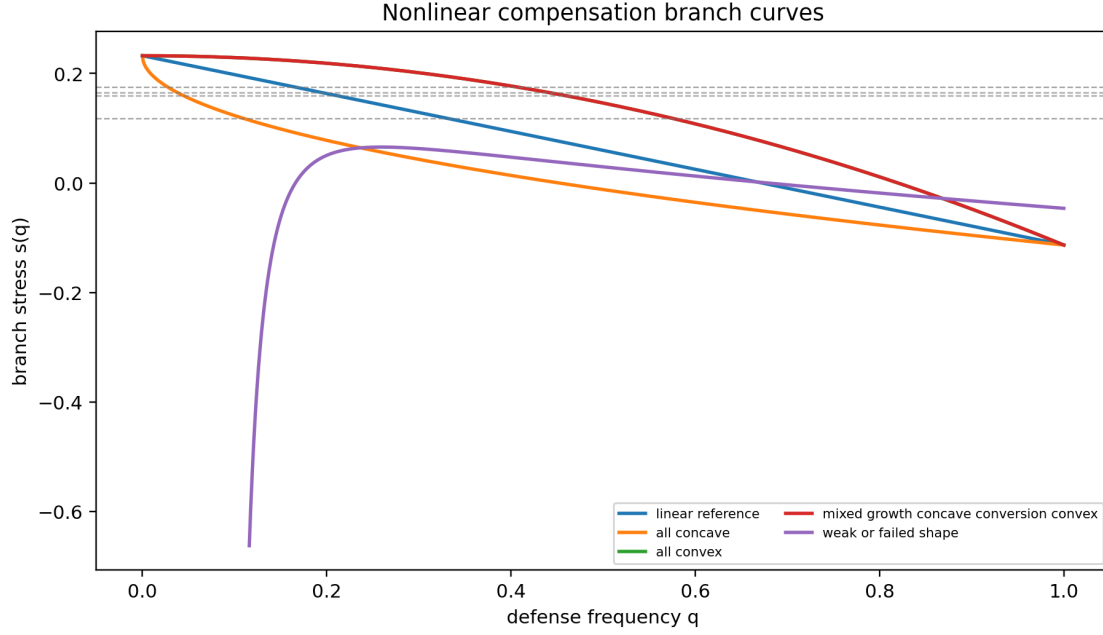


Figure 15: Generalized nonlinear compensation branch curves $s(q)$ for selected endpoint-preserving trade-off shapes. Target stresses are marked for comparison.

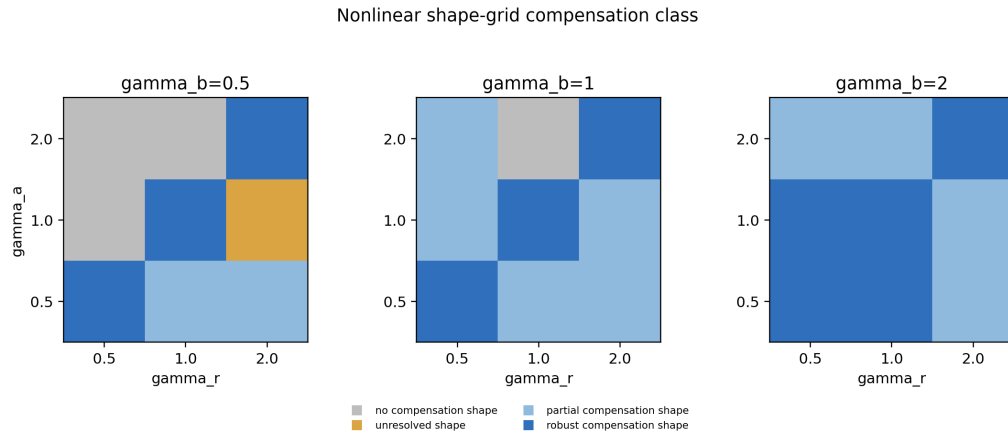


Figure 16: Controlled nonlinear shape-grid classification across $\gamma_r, \gamma_a, \gamma_b \in \{0.5, 1, 2\}$. The grid is structured and local, not a broad nonlinear trade-off scan.

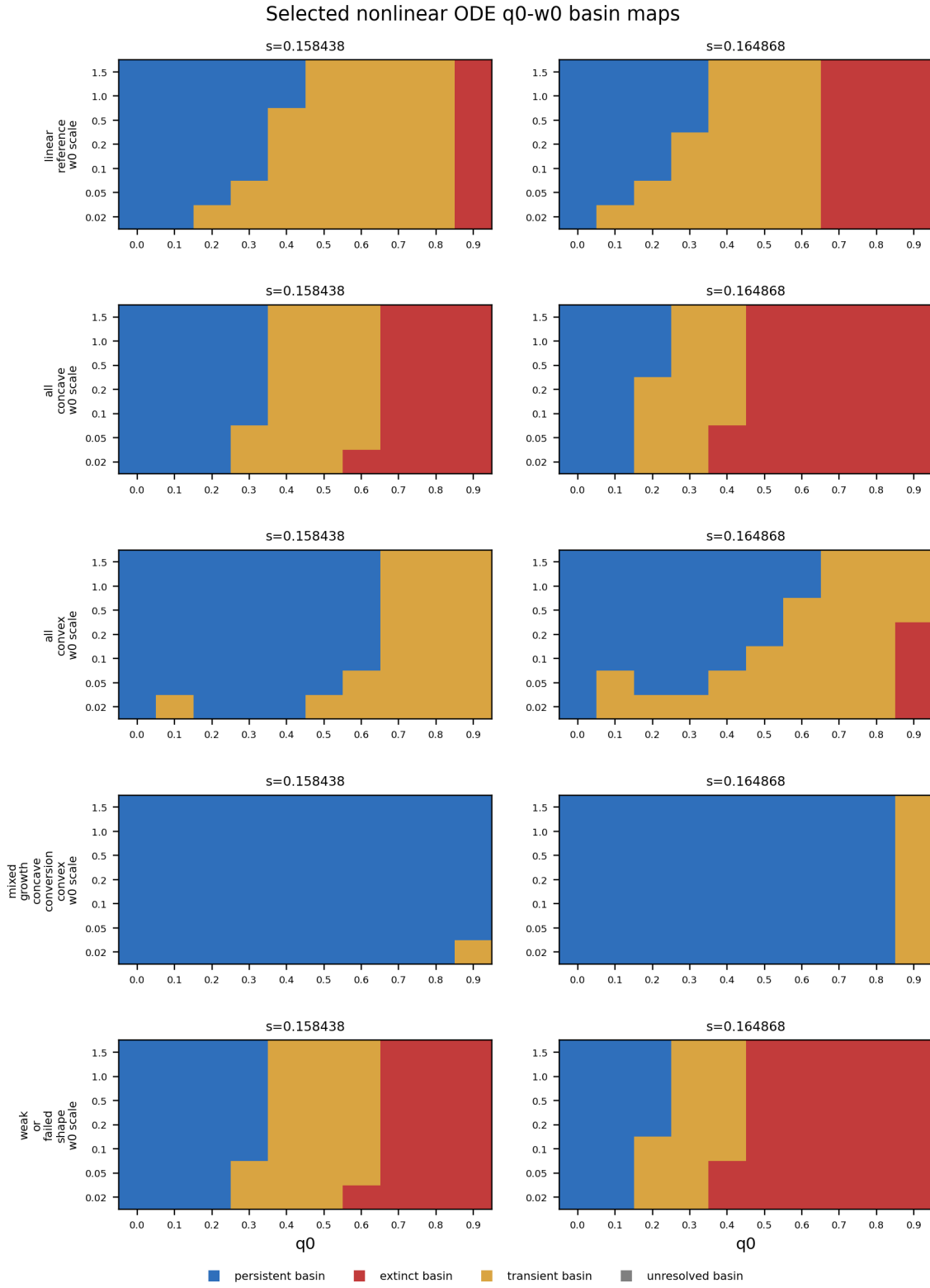


Figure 17: Selected homogeneous ODE basin maps for linear, concave, convex, mixed, and weak/failed nonlinear shape cases.

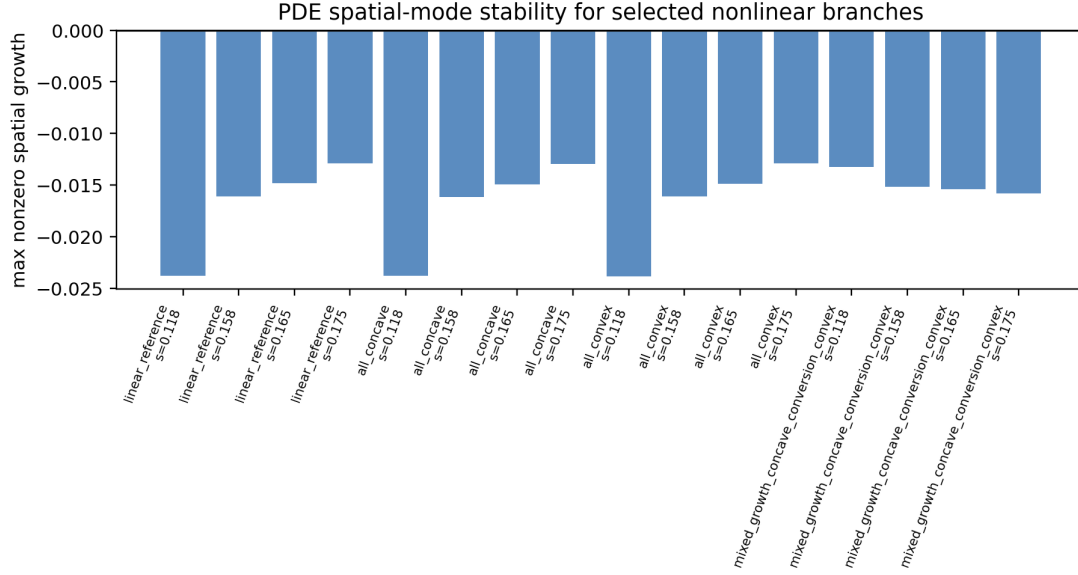


Figure 18: Maximum nonzero Neumann spatial-mode growth for selected nonlinear branch states. No selected stable branch case showed positive nonzero spatial growth.

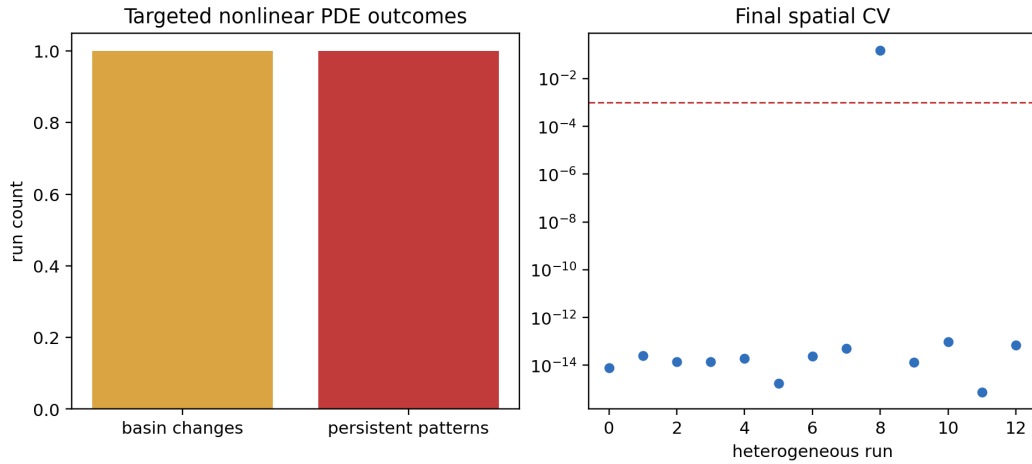


Figure 19: Targeted nonlinear PDE non-homogeneous test summary. The finite-horizon all-convex boundary transient resolved at longer horizon and did not support persistent spatial-pattern-mediated rescue.

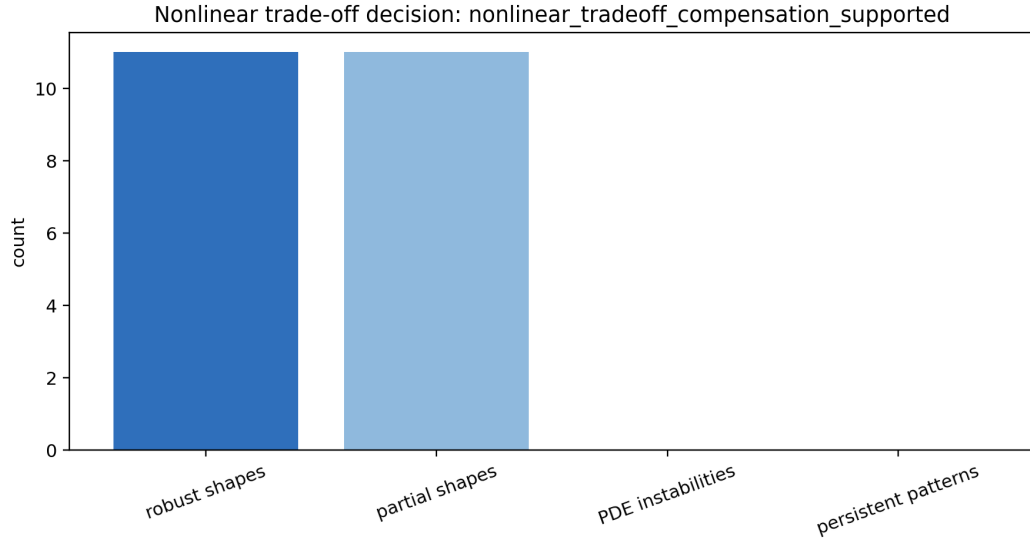


Figure 20: Final nonlinear trade-off extension decision summary. The supported label is conditional on the controlled shape grid and targeted PDE checks.

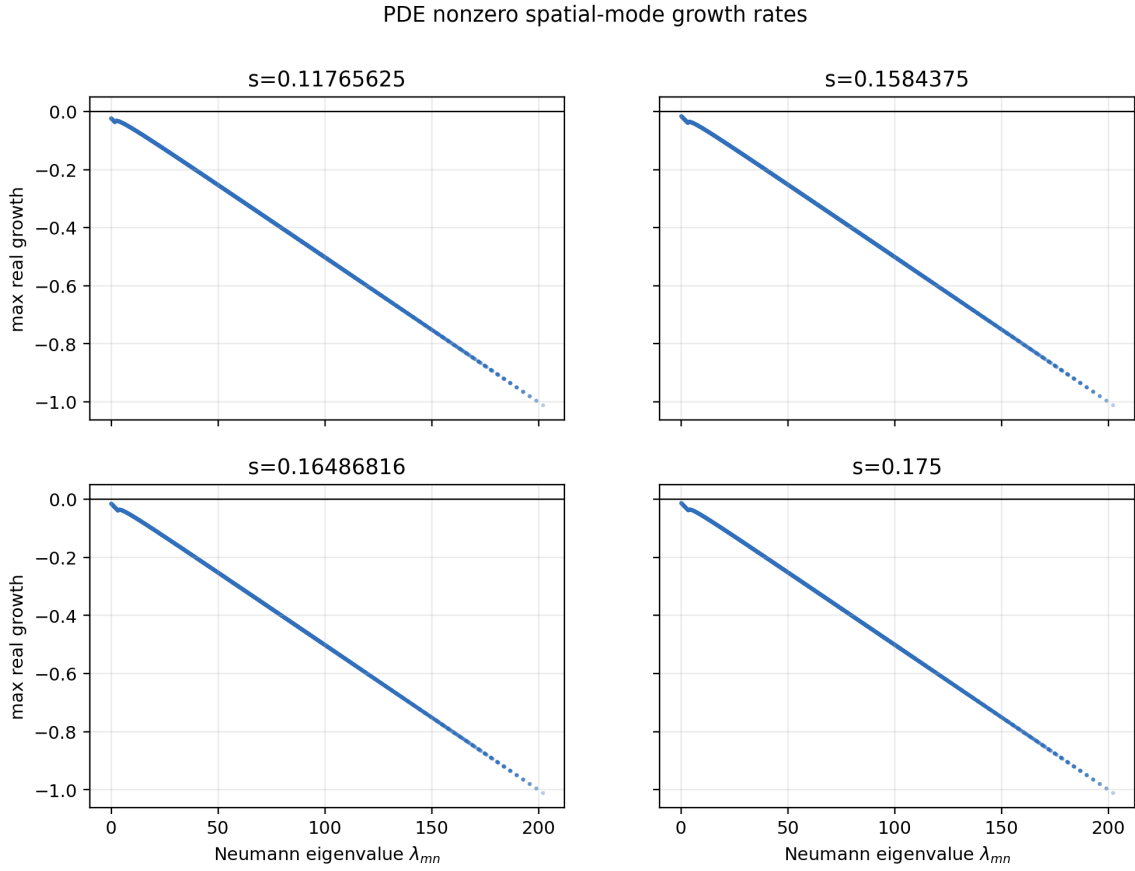


Figure 21: Nonzero Neumann spatial-mode growth rates along the homogeneous compensation branch under the current diffusion coefficients.

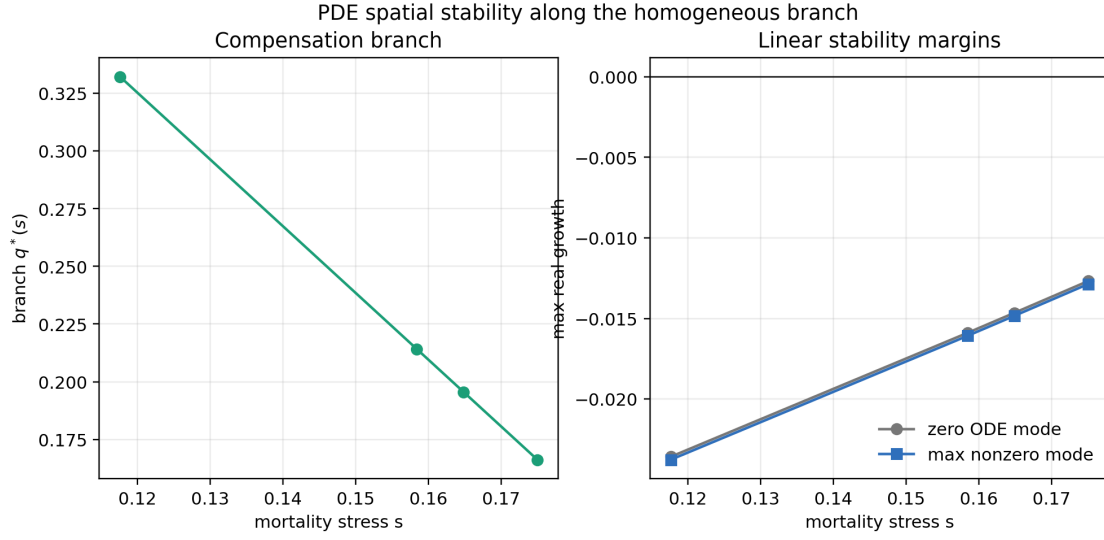


Figure 22: Spatial stability summary along the compensation branch. The largest nonzero mode growth rate remains negative at the tested stresses.

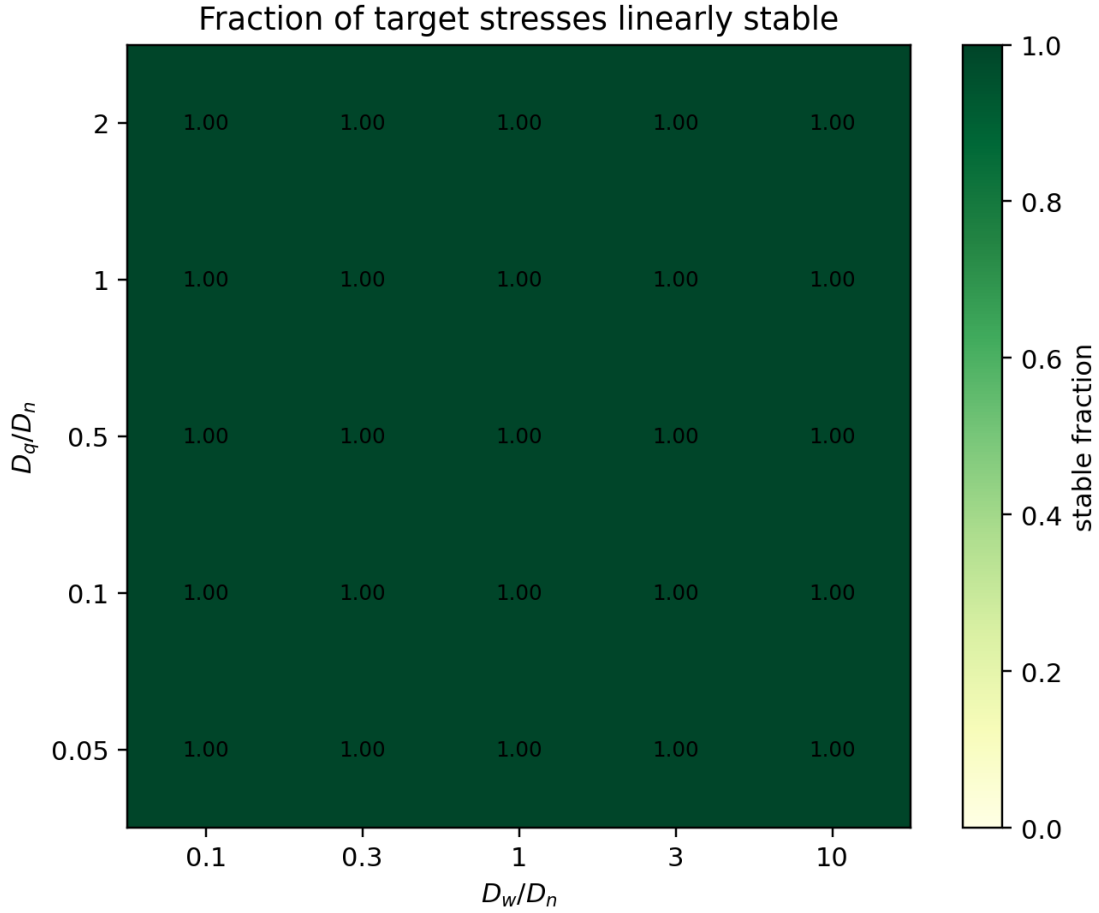


Figure 23: Controlled diffusion-ratio linear stability grid. The grid varies diffusion ratios only for the linear mode analysis and is not a nonlinear PDE parameter scan.

Linear PDE spatial-mode stability test

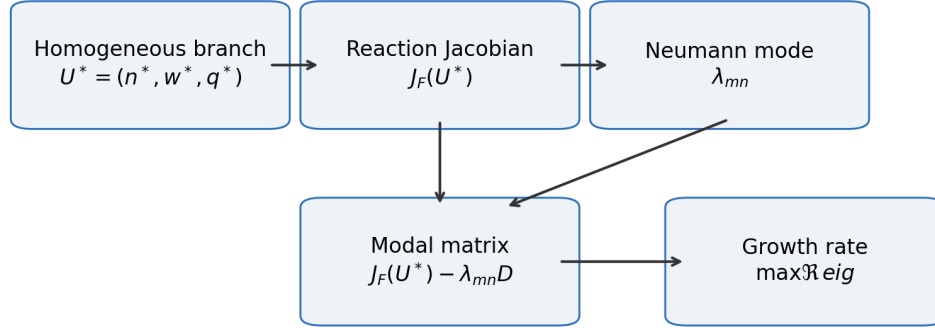


Figure 24: Schematic of the spatial-mode stability calculation for perturbations around the homogeneous compensation branch.

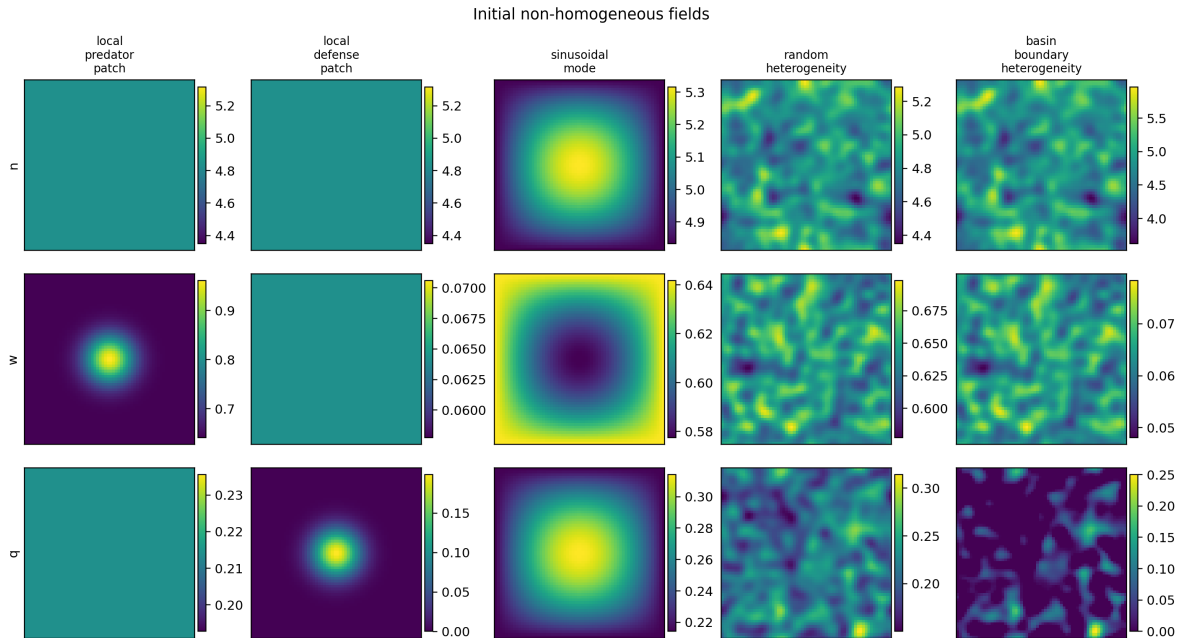


Figure 25: Representative non-homogeneous initial conditions used in the targeted nonlinear PDE perturbation tests.

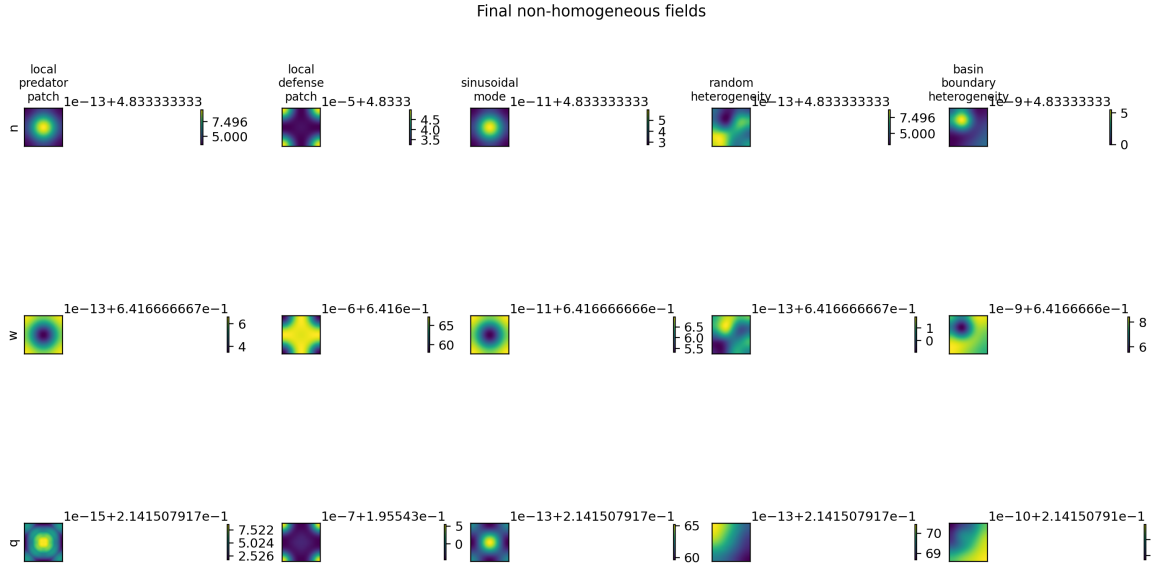


Figure 26: Final fields from representative non-homogeneous perturbation tests. Spatial variation decays in the representative saved fields.

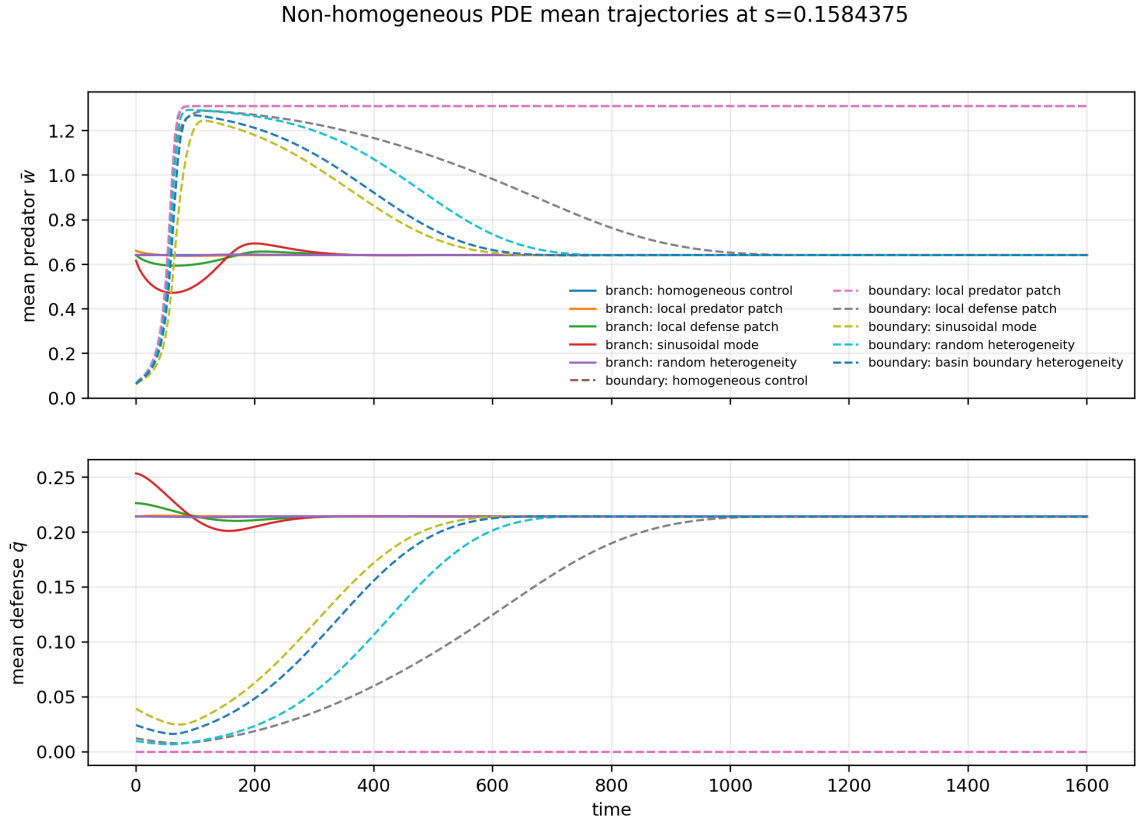


Figure 27: Spatial mean trajectories for homogeneous controls and selected non-homogeneous PDE perturbation tests.

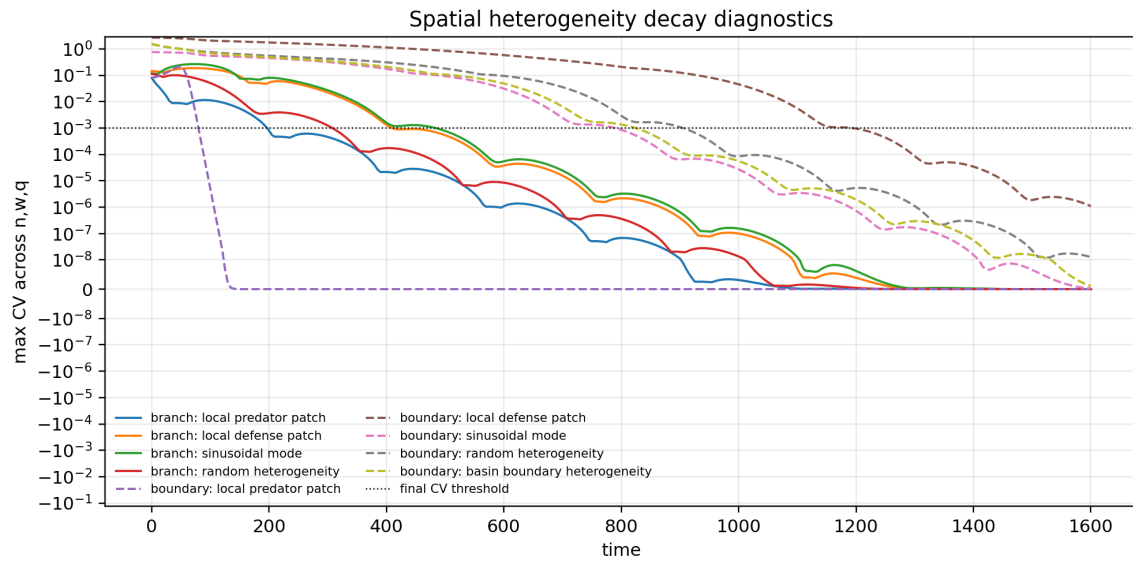


Figure 28: Spatial coefficient-of-variation diagnostics for selected non-homogeneous PDE perturbation tests.

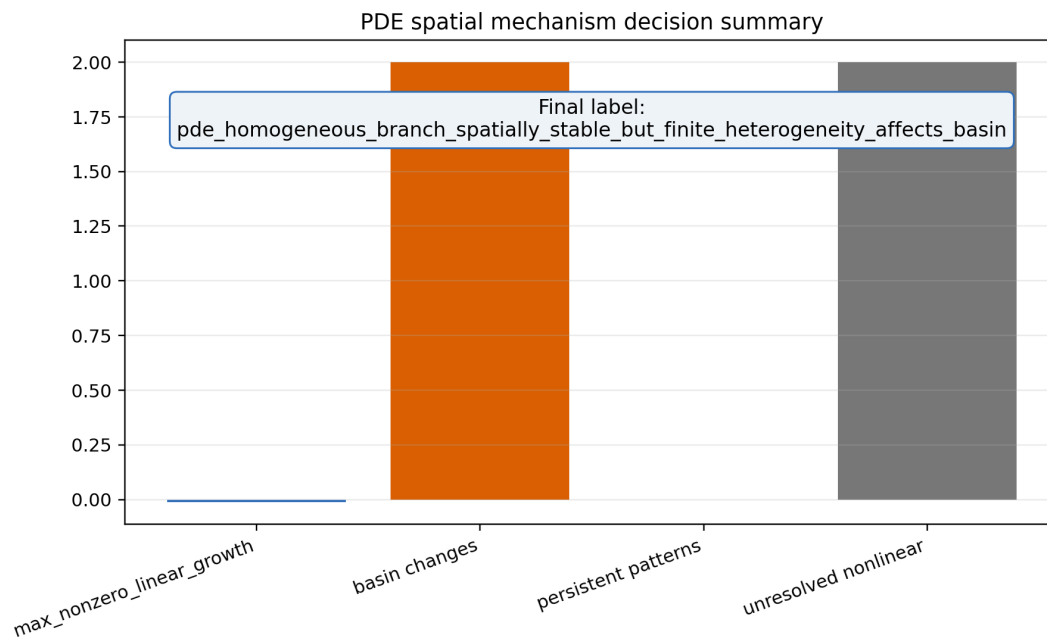


Figure 29: Decision summary for the PDE spatial-mode and finite-amplitude non-homogeneous perturbation analysis.

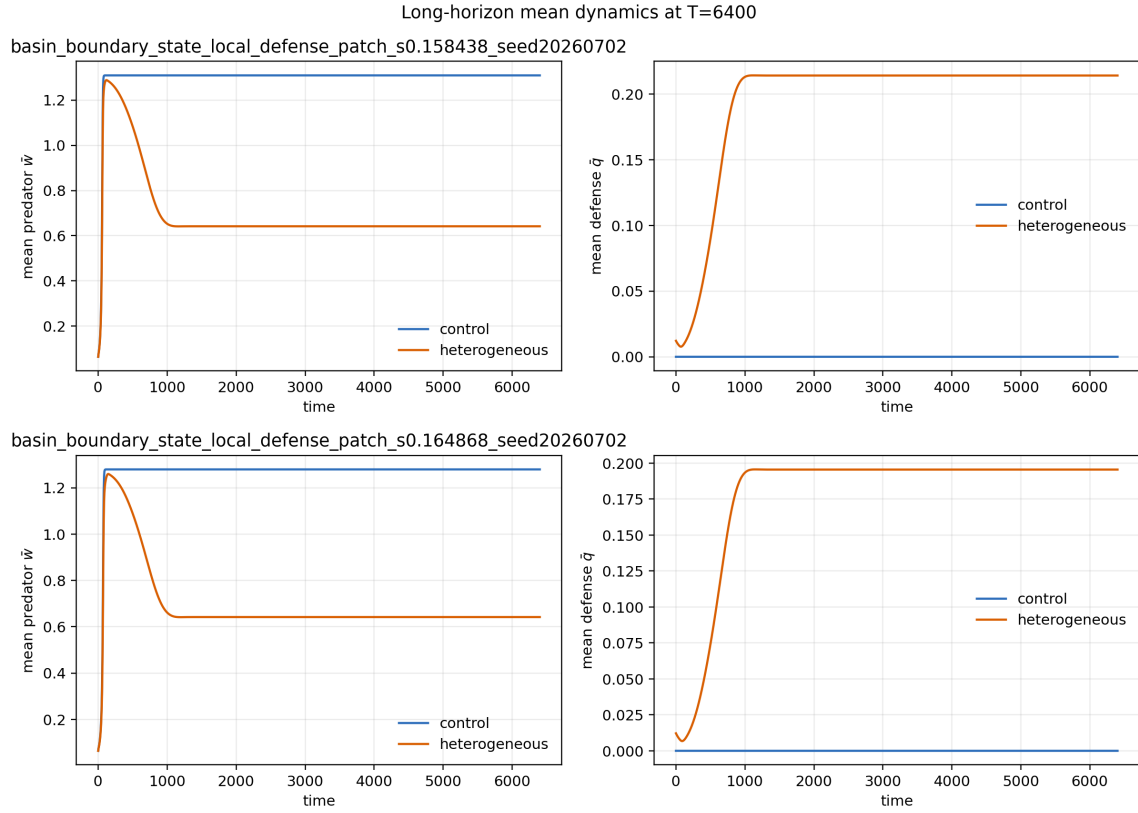


Figure 30: Long-horizon mean trajectories for the PR #23 basin-changing non-homogeneous cases and their matched homogeneous controls.

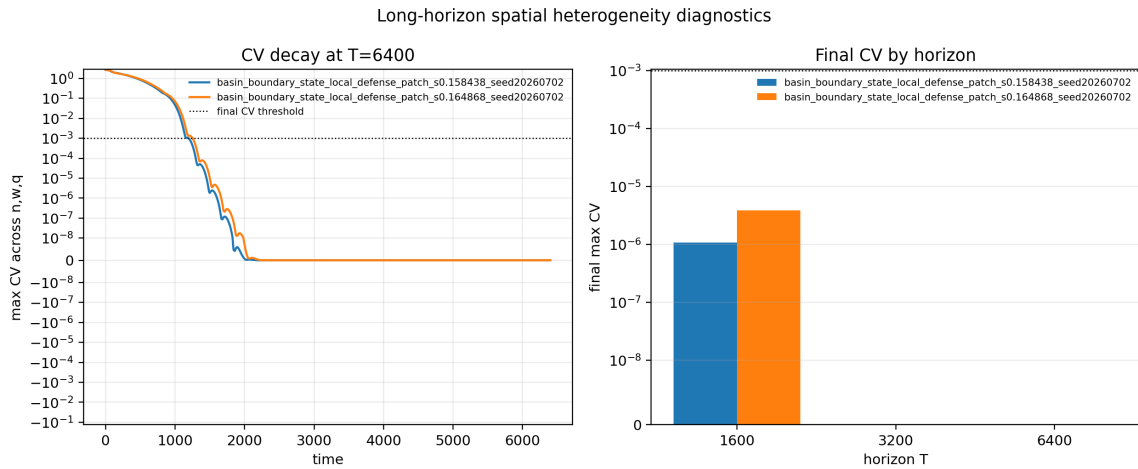


Figure 31: Spatial CV diagnostics for the long-horizon follow-up. Final spatial variation decays below the pattern-persistence threshold.

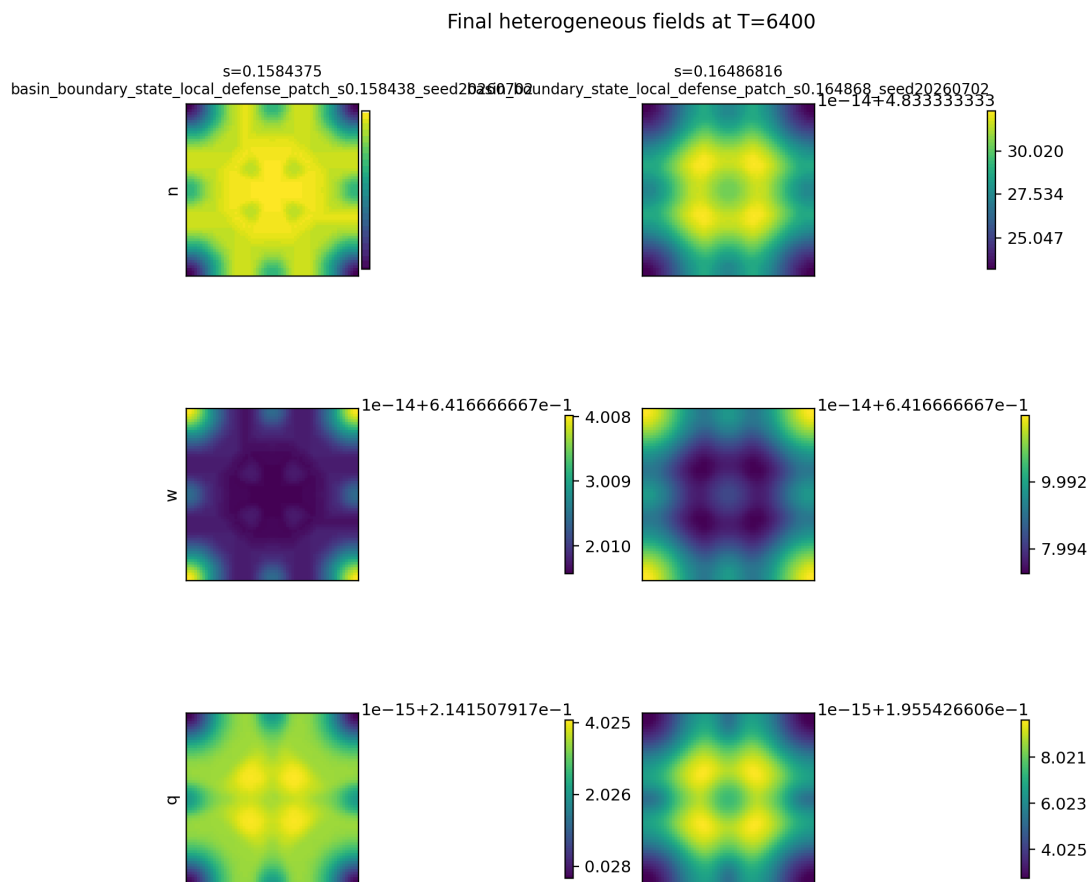


Figure 32: Final fields for the followed heterogeneous cases at the longest focused horizon.

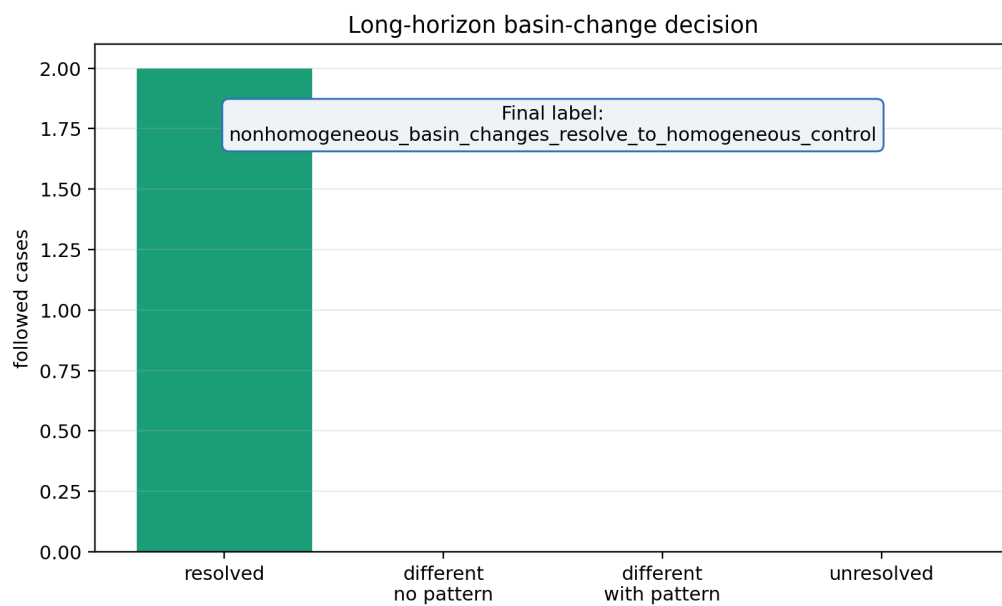


Figure 33: Decision summary for the long-horizon follow-up of the PR #23 basin-changing non-homogeneous cases.

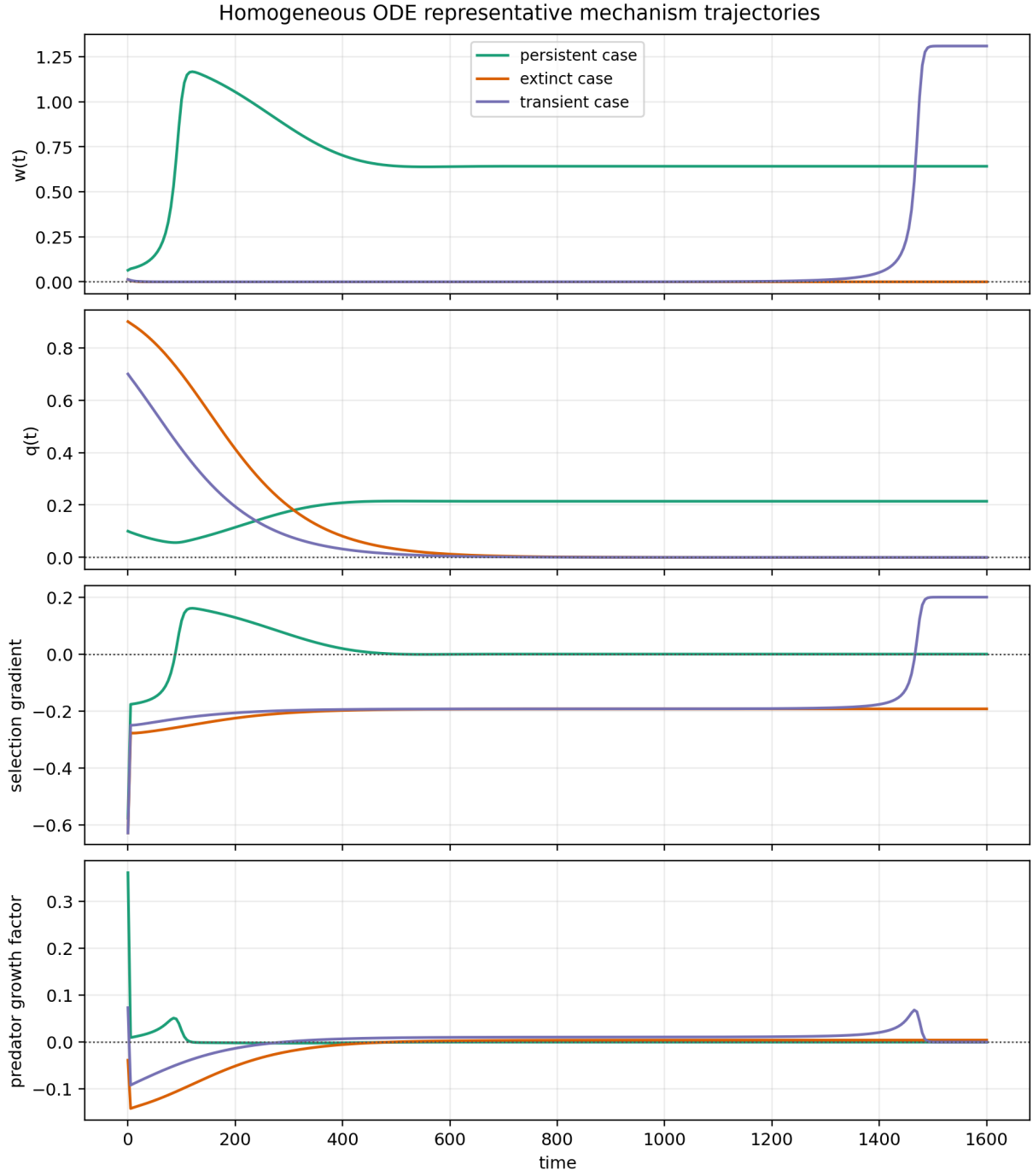


Figure 34: Representative homogeneous ODE trajectories for predator density w , defense frequency q , selection gradient, and predator growth factor. The cases use the same initial means as the representative PDE comparisons.

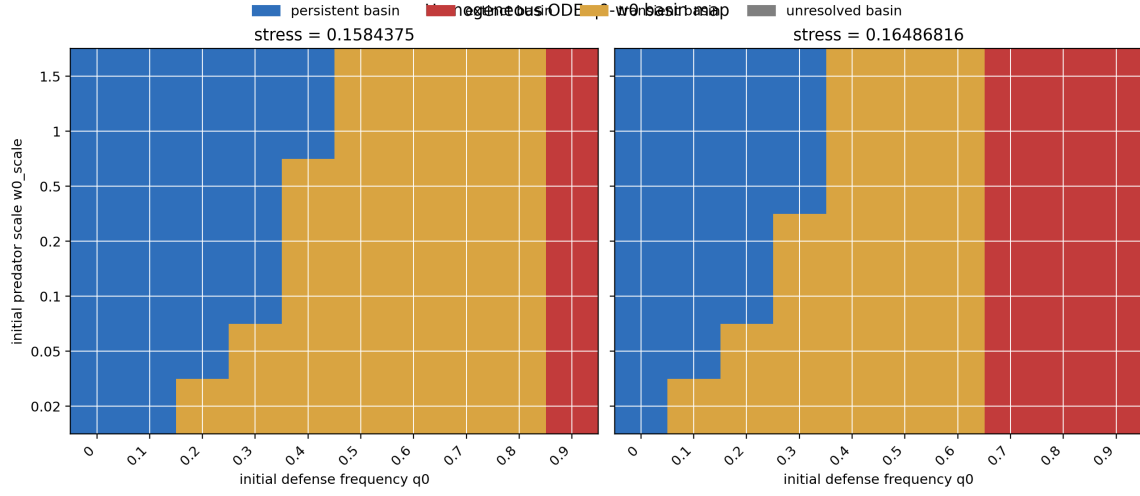


Figure 35: Homogeneous ODE basin map on the same q_0 - w_0 grid used in the ODE-PDE basin comparison.

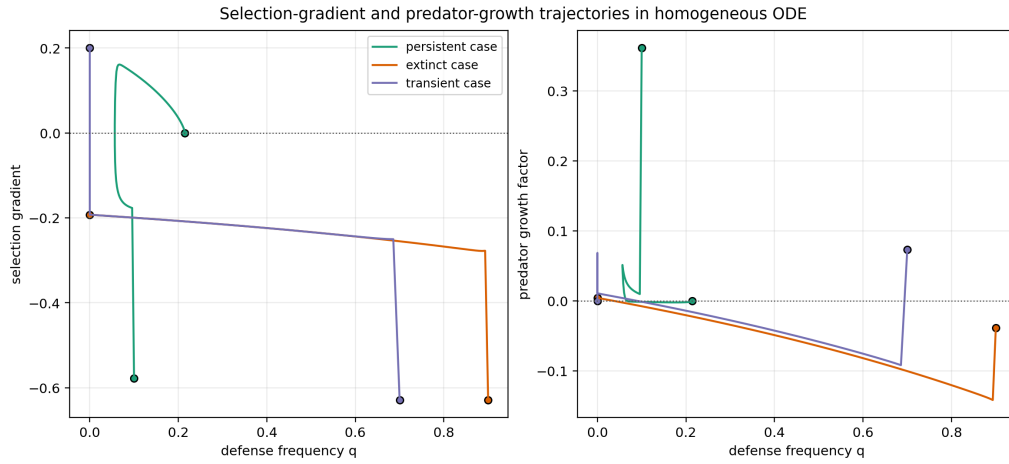


Figure 36: Representative ODE relationship between defense frequency q , the selection gradient on defense, and the predator growth factor.

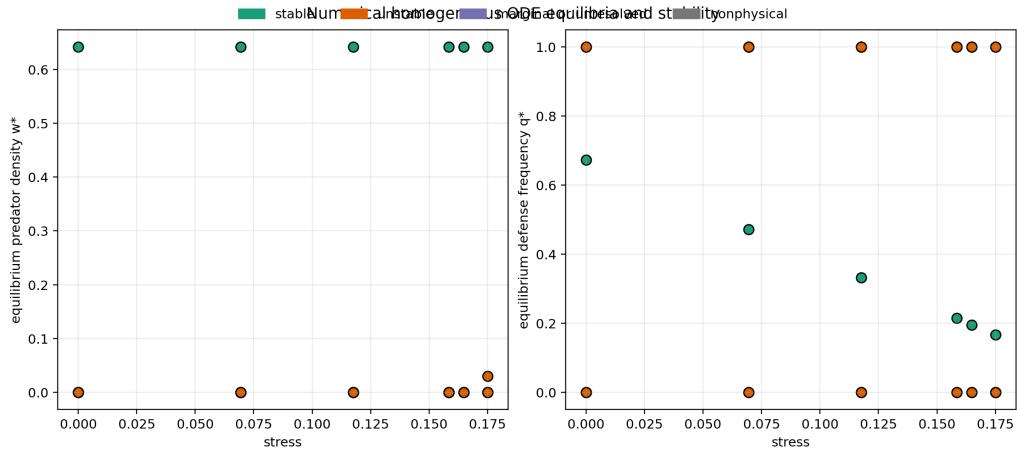


Figure 37: Numerical homogeneous equilibria across targeted stress values. Marker color indicates stability classification from finite-difference Jacobian eigenvalues.

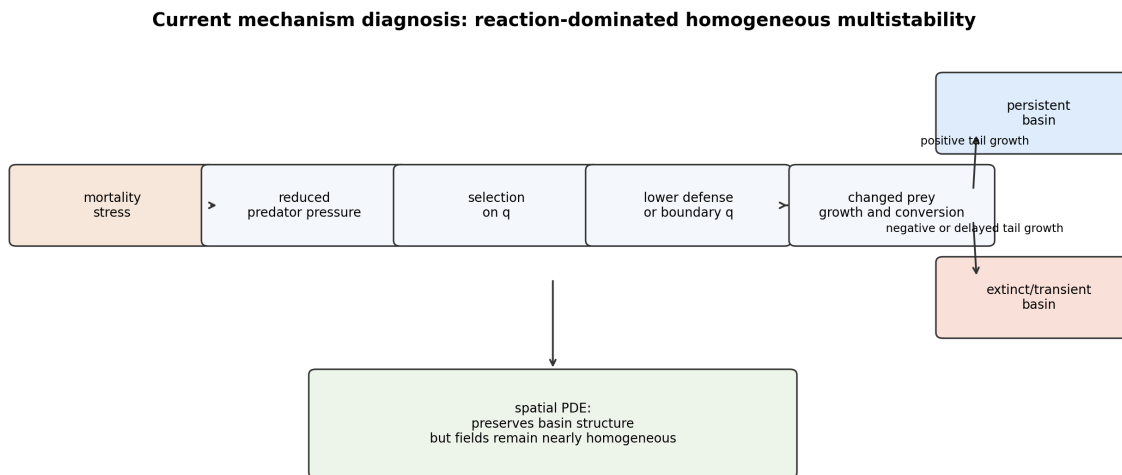


Figure 38: Updated mechanism diagram. Mortality stress alters predator pressure, which changes selection on prey defense and can move the homogeneous reaction system into predator-persistent, extinct, or transient basins. In the current tests, the spatial PDE preserves this basin structure while representative solutions remain nearly homogeneous.

10 Discussion

The fixed-defense control shows that spatial patterning alone was insufficient for robust predator rescue in the tested Roy-style model. The well-mixed eco-evolutionary model then shows that evolutionary change in prey defense can rescue predators under mortality stress, consistent with indirect evolutionary rescue theory [Yamamichi and Miner, 2015]. The spatially extended eco-evolutionary model preserves basin-dependent outcomes, but the current mechanism audit indicates that those basins are primarily inherited from homogeneous reaction dynamics.

The PDE result is not a simple amplification or suppression of the well-mixed rescue threshold; it is a spatially extended expression of basin-dependent dynamics. This changes what “spatial effect on rescue” means. Rather than asking whether the PDE threshold is higher or lower than the ODE threshold, the relevant question is which outcomes are reachable at each stress, how the basins depend on initial ecological and evolutionary state, and whether those basins are generated by homogeneous reactions or by persistent spatial patterning.

The basin-boundary results indicate that initial defense frequency and predator abundance are important coordinates for basin entry. Low-to-intermediate defense frequencies more often reach predator persistence, while high-defense states and some low-predator states more often lead to extinction or unresolved transients. Because transient outcomes remain common, the current basin map should be read as a resolved first layer of basin structure, not as a fully refined separatrix.

10.1 Strength of current claims

Strongly supported in the tested parameterization:

- well-mixed prey defense evolution increases predator mortality tolerance;
- fixed-defense spatial patterning did not produce robust predator rescue;
- the spatially extended eco-evolutionary model preserves basin-dependent outcomes in the tested stress range;
- the current mechanism diagnosis is homogeneous compensation preserved by the PDE rather than spatial-pattern-mediated rescue.

Moderately supported:

- initial defense frequency q_0 and predator abundance w_0 influence basin entry.
- finite-amplitude local-defense heterogeneity can shift finite-horizon basin assignment near tested basin-boundary states, but the followed cases resolve to the homogeneous-control basin by longer horizons.

Unresolved:

- the exact separatrix in q_0 – w_0 space;
- nonlinear basin sensitivity across diffusion coefficients beyond the current targeted perturbation tests;
- generality across trade-off forms and parameterizations;
- global basin geometry of the homogeneous reaction system;
- biological calibration.

11 Limitations and future work

The current analysis has several limitations. It uses one baseline trade-off parameterization, with $b_u = 0.08$ and $b_v = 0.02$, plus a small controlled nonlinear shape grid. Transient outcomes remain common near the basin boundary. The q_0 – w_0 basin boundary has not yet been adaptively refined. The diffusion-ratio analysis is linear, and the nonlinear spatial perturbation tests are targeted rather than exhaustive. The model has also not been externally calibrated to a particular biological system.

The next numerical task, if needed, is locally refined basin-boundary diagnosis rather than broad PDE scanning. Further PDE basin refinement may be useful later, but it should follow the current mechanism correction rather than replace it.

In the well-mixed system, prey defense evolution supports indirect evolutionary rescue through a homogeneous compensation branch. In the spatially extended system, the same eco-evolutionary reaction mechanism does not reduce to a single persistence threshold. Instead, the PDE preserves reaction-level basin structure, with basin entry controlled in part by initial prey defense frequency and predator abundance. After the long-horizon follow-up of the non-homogeneous basin-changing cases, the apparent basin changes caused by local-defense heterogeneity resolved to the matched homogeneous-control basin. The controlled nonlinear trade-off extension recovers the linear branch and supports selected nonlinear compensation branches, while showing that branch feasibility and stability depend on trade-off shape. Together, these results support the interpretation that the tested PDE preserves the homogeneous compensation mechanism rather than producing persistent spatial-pattern-mediated rescue.

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